MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

FLIP-FLOP, JK, DIGITAL, KEY CODE EBC

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic JK flip-flop module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.6 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be EBC.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/172-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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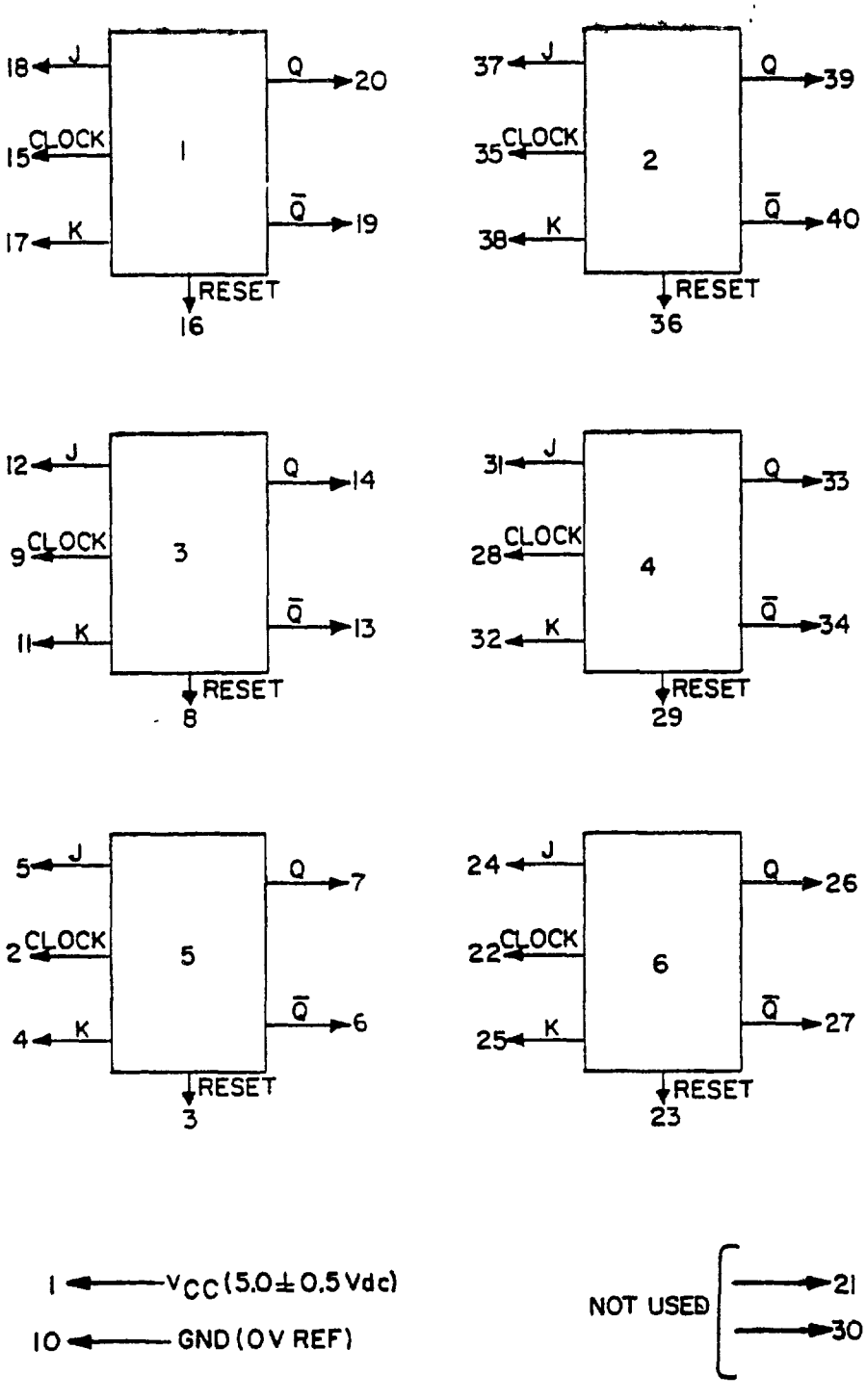


FIGURE 1. EBC module function diagram, JK flip-flop.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A12	V_{OUT} (low-loaded)	III	S	0.0	0.6	0.0	0.6	0.0	0.6	0.0	0.6	V
B1-B6	Input breakdown current	IV	S	0	200	0	200	0	200	0	200	µA
B7-B18	Input breakdown current	IV	S	0	100	0	100	0	100	0	100	µA
B19-B24	Input breakdown current	IV	S	0	200	0	200	0	200	0	200	µA
C1	Isolation, frame	V	S	10	∞	1	∞	10	∞	1	∞	MΩ
C2-C3	Isolation, unused pins	V	S	10	∞	1	∞	10	∞	1	∞	MΩ
D1-D6	Input current, high level	VI	X	0	-200	0	-200	0	-200	0	-200	µA
D7-D18	Input current, high level	VI	X	0	10	0	10	0	10	0	10	µA
D19-D24	Input current, high level	VI	X	0	20	0	20	0	20	0	20	µA
E1-E6	Input current, low level	VI	X	10	600	10	600	10	600	10	600	µA
E7-E18	Input current, low level	VI	X	10	200	10	200	10	200	10	200	µA
E19-E24	Input current, low level	VI	X	10	400	10	400	10	400	10	400	µA
F1-F12	Output short circuit current	VII	X	3	15	3	15	3	15	3	15	mA
G1-G6	Output voltage, high level (J threshold high)	VIII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
H1-H6	Output voltage, high level (K threshold high)	VIII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J6	Output voltage, high level (Reset threshold low)	VIII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
J7-J12	Output voltage, high level (Reset threshold high)	VIII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
K1-K6	Output voltage, low level (J threshold low)	VIII	X	0.0	0.3	0.0	0.3	0.0	0.3	0.0	0.3	V
K7-K12	Output voltage, low level (K threshold low)	VIII	X	0.0	0.3	0.0	0.3	0.0	0.3	0.0	0.3	V
L	Power supply current (Q outputs high level)	IX	S	2.0	8.6	2.0	8.6	2.0	8.6	2.0	8.6	mA
M1-M6	JK operation (J and K threshold high) (max freq, min pulse width)	X	S	760	840	760	840	760	840	760	840	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
M7-M12	JK operation (J and K threshold low) (max freq, min pulse width)	X	S	0.0	0.3	0.0	0.3	0.0	0.3	0.0	0.3	V
N1-N12	Transition time (t_{TLH}), C_L = 40 pF	XI	S	5	60	5	60	5	60	5	60	ns
N13-N24	Transition time (t_{TLH}), C_L = 140 pF	XI	Q	10	100	10	100	10	100	10	100	ns
P1-P12	Transition time (t_{THL}), C_L = 40 pF	XI	S	2	30	2	30	2	30	2	30	ns
P13-P24	Transition time (t_{THL}), C_L = 140 pF	XI	Q	5	60	5	60	5	60	5	60	ns
Q1-Q12	Propagation delay time (t_{PLH}), C_L = 40 pF	XI	S	10	75	10	75	10	90	10	90	ns
Q13-Q24	Propagation delay time (t_{PLH}), C_L = 140 pF	XI	Q	10	105	10	105	10	120	10	120	ns
R1-R12	Propagation delay time (t_{PHL}), C_L = 40 pF	XI	S	10	150	10	150	10	165	10	165	ns
R13-R24	Propagation delay time (t_{PHL}), C_L = 140 pF	XI	Q	10	180	10	180	10	195	10	195	ns
S1-S6	Propagation delay time (t_{PHL}), (V_{IN} clock = 0 V), C_L = 40 pF	XII	S	10	200	10	200	20	220	10	220	ns
S7-S12	Propagation delay time (t_{PLH}), (V_{IN} clock = 2.4 V), C_L = 40 pF	XII	S	10	75	10	75	10	90	10	90	ns
S13-S18	Propagation delay time (t_{PHL}), (V_{IN} clock = 0 V), C_L = 140 pF	XII	Q	10	230	10	230	10	250	10	250	ns
S19-S24	Propagation delay time (t_{PLH}), (V_{IN} clock = 2.4 V), C_L = 140 pF	XII	S	10	105	10	105	10	120	10	120	ns
T1-T6	Propagation delay time (t_{PLH}), (V_{IN} clock = 0 V), C_L = 40 pF	XII	S	10	75	10	75	10	90	10	90	ns
T7-T12	Propagation delay time (t_{PHL}), (V_{IN} clock = 2.4 V), C_L = 40 pF	XII	S	10	150	10	150	10	165	10	165	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
T13-T18	Propagation delay time (t_{PLH}), (V_{IN} clock = 0 V), C_L = 140 pF	XII	Q	10	105	10	105	10	115	10	115	ns
T19-T24	Propagation delay time (t_{PHL}), (V_{IN} clock = 2.4 V), C_L = 140 pF	XII	Q	10	180	10	180	10	195	10	195	ns
U1-U6	Maximum clock fall time	XIII	S	5.7	6.3	5.7	6.3	5.7	6.3	5.7	6.3	µs
V1-V6	Threshold clock input	XIII	S	0.7	0.9	0.7	0.9	0.7	0.9	0.7	0.9	µs
W1-W12	Output crosstalk, low level	XIV	Q	0	400	0	400	0	400	0	400	mV
X1-X12	Output crosstalk, high level	XIV	Q	0	400	0	400	0	400	0	400	mV
Y	Power supply transient crosstalk	XV	Q	0	400	0	400	0	400	0	400	mV
Z	Power supply filter capacitance	XVI	S	1.65	2.75	1.65	2.75	1.65	2.75	1.65	2.75	µF
AA1-AA24	Input capacitance	XVI	Q	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through AA shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through AA in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through Z shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through Z shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through Z shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Part number M28787/172-1 supersedes part number M28787/172. The two parts are interchangeable.

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* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation		48 mW (outputs not loaded)
Supply voltage		5 Vdc ±10 percent
Input voltage, J, K, reset		
Low level		-0.5 V minimum, 0.7 V maximum
High level		2.0 V minimum, V_{CC} maximum
Clock input voltage		
Low level		-0.5 V minimum, 0.6 V maximum
High level		2.0 V minimum, V_{CC} maximum
Input current low level		
Clock	(V_{IL} = 0.0 V)	-0.60 mA maximum
	(V_{IL} = 0.3 V)	-0.36 mA maximum
J and K	(V_{IL} = 0.0 V)	-0.20 mA maximum
	(V_{IL} = 0.3 V)	-0.18 mA maximum
Reset	(V_{IL} = 0.0 V)	-0.40 mA maximum
	(V_{IL} = 0.3 V)	-0.36 mA maximum
Input current high level		
Clock	(V_{IH} = 2.4 V)	0 µA minimum, -200 µA maximum
	(V_{IH} = 5.5 V)	200 µA maximum
J and K	(V_{IH} = 2.4 V)	10 µA maximum
	(V_{IH} = 5.5 V)	100 µA maximum
Reset	(V_{IH} = 2.4 V)	20 µA maximum
	(V_{IH} = 5.5 V)	200 µA maximum
Output voltage		
Low level	(I_{OL} = 2.0 mA)	0.3 V maximum
High level	(I_{OH} = -100 µA)	2.4 V minimum
Output drive current maximum capability		
Low level	(V_{OL} = 0.3 V)	2.0 mA
High level	(V_{OH} = 2.4 V)	-100 µA
Clock input		
Frequency		2.5 MHz maximum
Pulse width		200 ns minimum
Fall time		1.0 µs maximum
Input setup time		≥ clock pulse width
Input hold time		0 ns minimum
Reset pulse width		100 ns minimum

* 6.5 Module function. This module contains six indepedent JK flip-flops with both true (Q) and complementary ($\overline{Q}$) outputs as illustrated on figure 1. Each JK flip-flop has a J, K, clock, and reset input. Information is transferred into a temporary memory when the clock goes to a high state. When the clock returns low, the information is transferred to the Q and $\overline{Q}$ outputs. The information at the J and K inputs should be present when the clock goes high and the information should not be changed while the clock is high. The application of a low state to the reset will cause the Q output to go to a low state. Reset inputs dominate regardless of the state of clock or logic inputs.

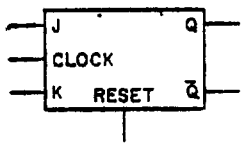
6.5.1 Input to output relationship. Each JK flip-flop has the input to output relationship of table II for positive logic.

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TABLE II. Input to output relationship. 1/-5/

Inputs				Outputs			
Reset	J	K	Clock	Qn	Qn+1	$\overline{Q}n$	$\overline{Q}n+1$
L	X	X	X	X	L	X	H
H	L	L	Pulse	L	L	H	H
H	L	L	Pulse	H	H	L	L
H	L	H	Pulse	L	L	H	H
H	L	H	Pulse	H	L	L	H
H	H	L	Pulse	L	H	H	L
H	H	L	Pulse	H	H	L	L
H	H	H	Pulse	L	H	H	L
H	H	H	Pulse	H	L	L	H

1/ L = low level.
2/ H = high level.
3/ X = Don't care.
4/ Qn, $\overline{Q}n$ = Output before clock pulse.
5/ Qn+1, $\overline{Q}n+1$ = Output after clock pulse.



6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

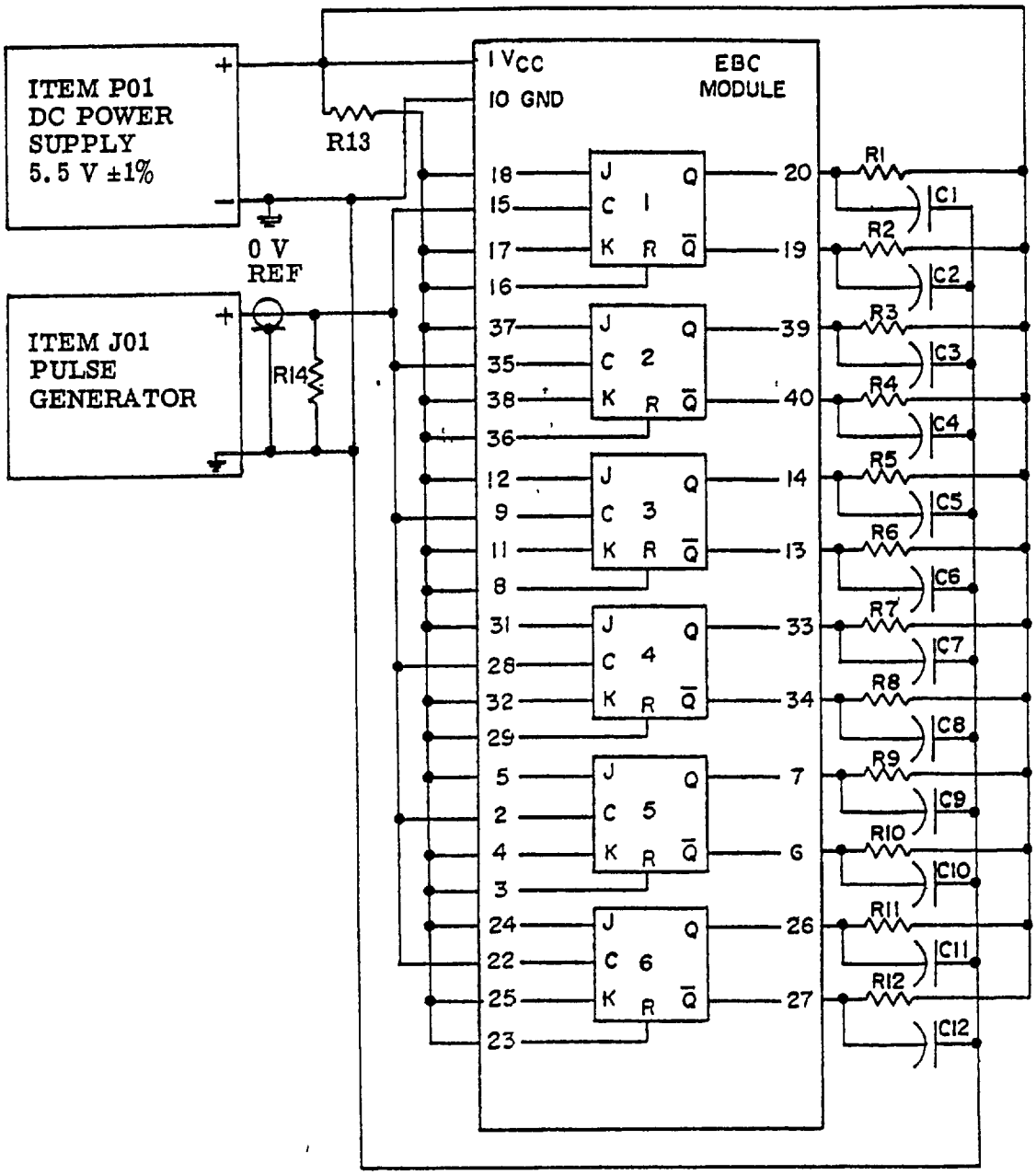
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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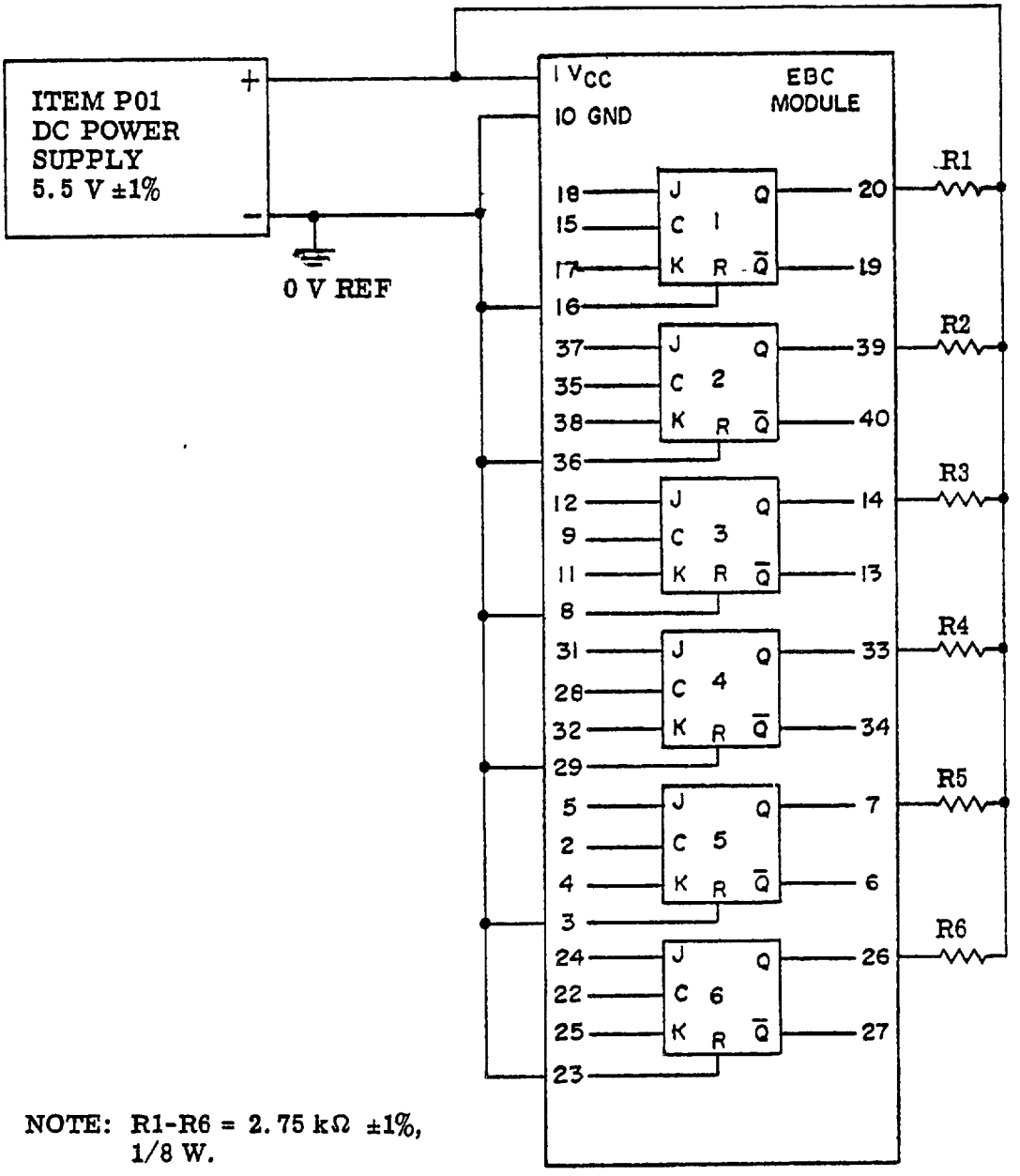
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NOTES:
1. R1-R12 = 2.75 kΩ ±1%, 1/8 W.
2. R13 = 1 kΩ ±1%, 1/8 W.
3. R14 = 50Ω ±5%, 1 W.
4. C1-C12 = 100 pF ±20%.
5. Item J01 setup:
   PA = 4 ±0.5 V.
   PW = 500 ±50 ns.
   $t_{TLH} = t_{THL}$ = 25 ±5 ns.
   PRR = 1.0 ±0.1 MHz.

FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1-R6 = 2.75 kΩ ±1%, 1/8 W.

FIGURE 3. Component temperature test circuit.

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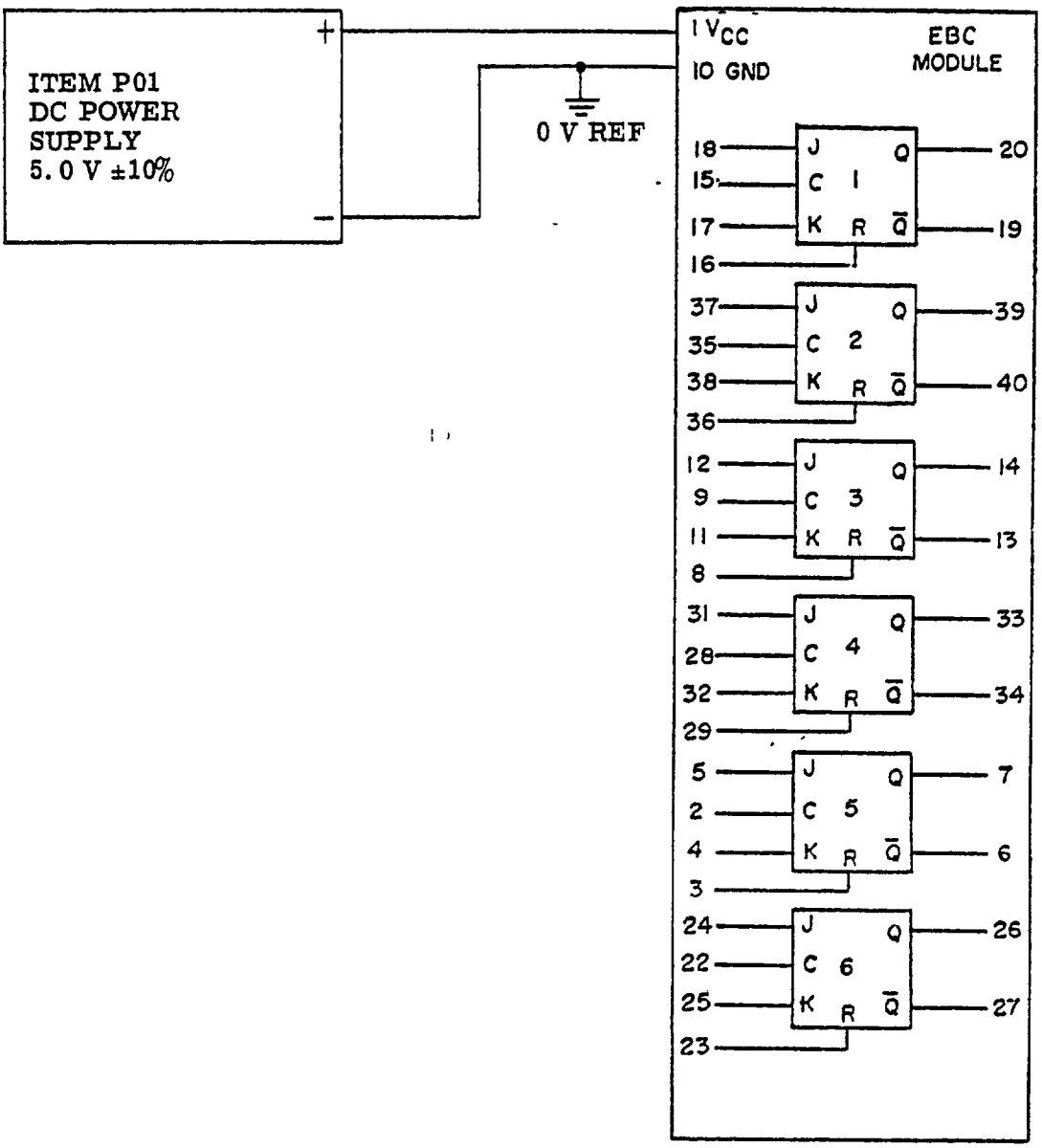


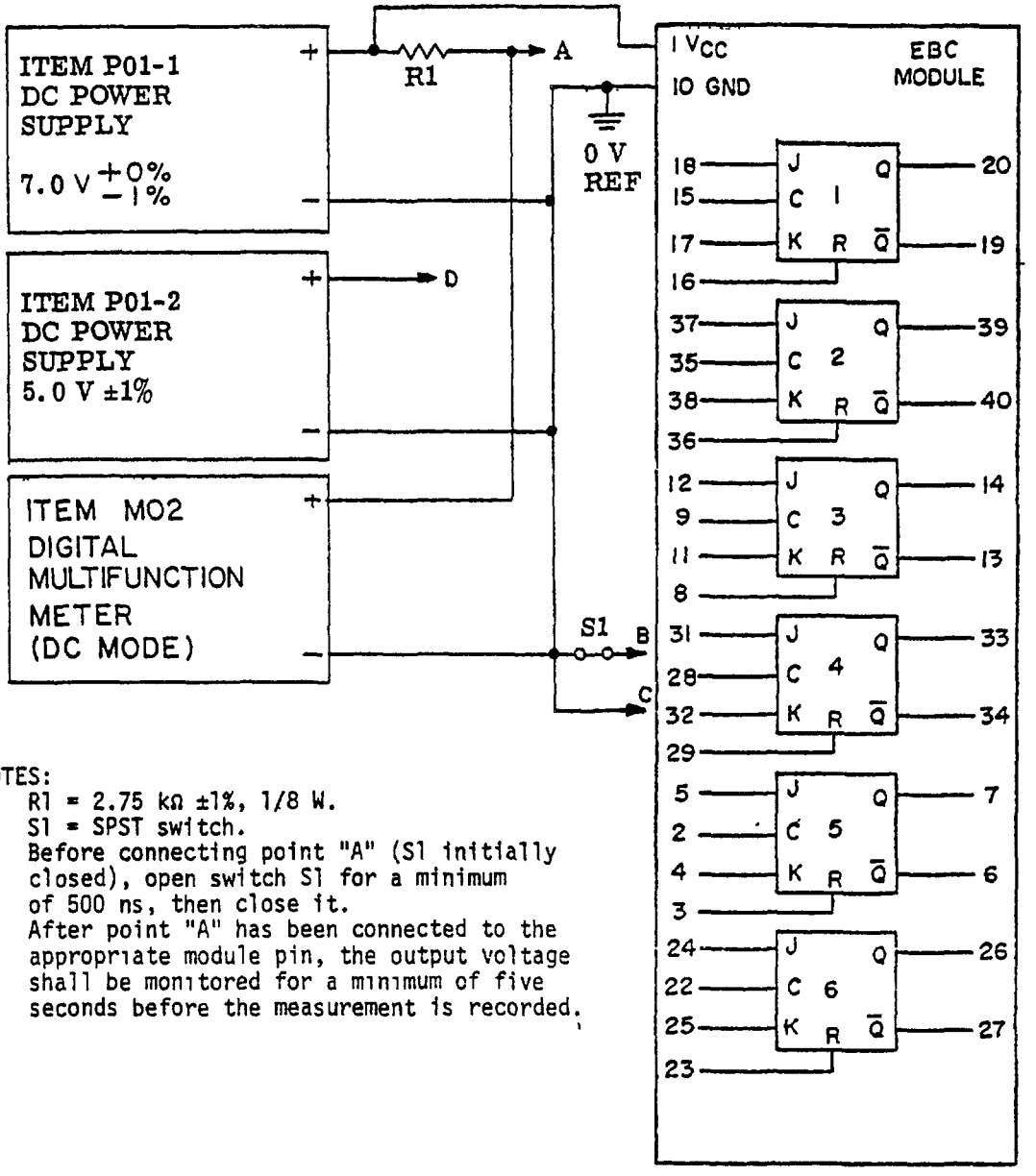
FIGURE 4. Vibration test circuit.

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TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
A1	$\overline{Q}1$ V_{OUT} (low loaded)	19	15	17	16	0.0	0.6	V	5
A2	$\overline{Q}2$	40	35	38	36				
A3	$\overline{Q}3$	13	9	11	8				
A4	$\overline{Q}4$	34	28	32	29				
A5	$\overline{Q}5$	6	2	4	3				
A6	$\overline{Q}6$	27	22	25	23				
A7	Q1	20	Open	16	Open				
A8	Q2	39		36					
A9	Q3	14		8					
A10	Q4	33		29					
A11	Q5	7		3					
A12	Q6	26		23					

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NOTES:
1. R1 = 2.75 kΩ ±1%, 1/8 W.
2. S1 = SPST switch.
3. Before connecting point "A" (S1 initially closed), open switch S1 for a minimum of 500 ns, then close it.
4. After point "A" has been connected to the appropriate module pin, the output voltage shall be monitored for a minimum of five seconds before the measurement is recorded.

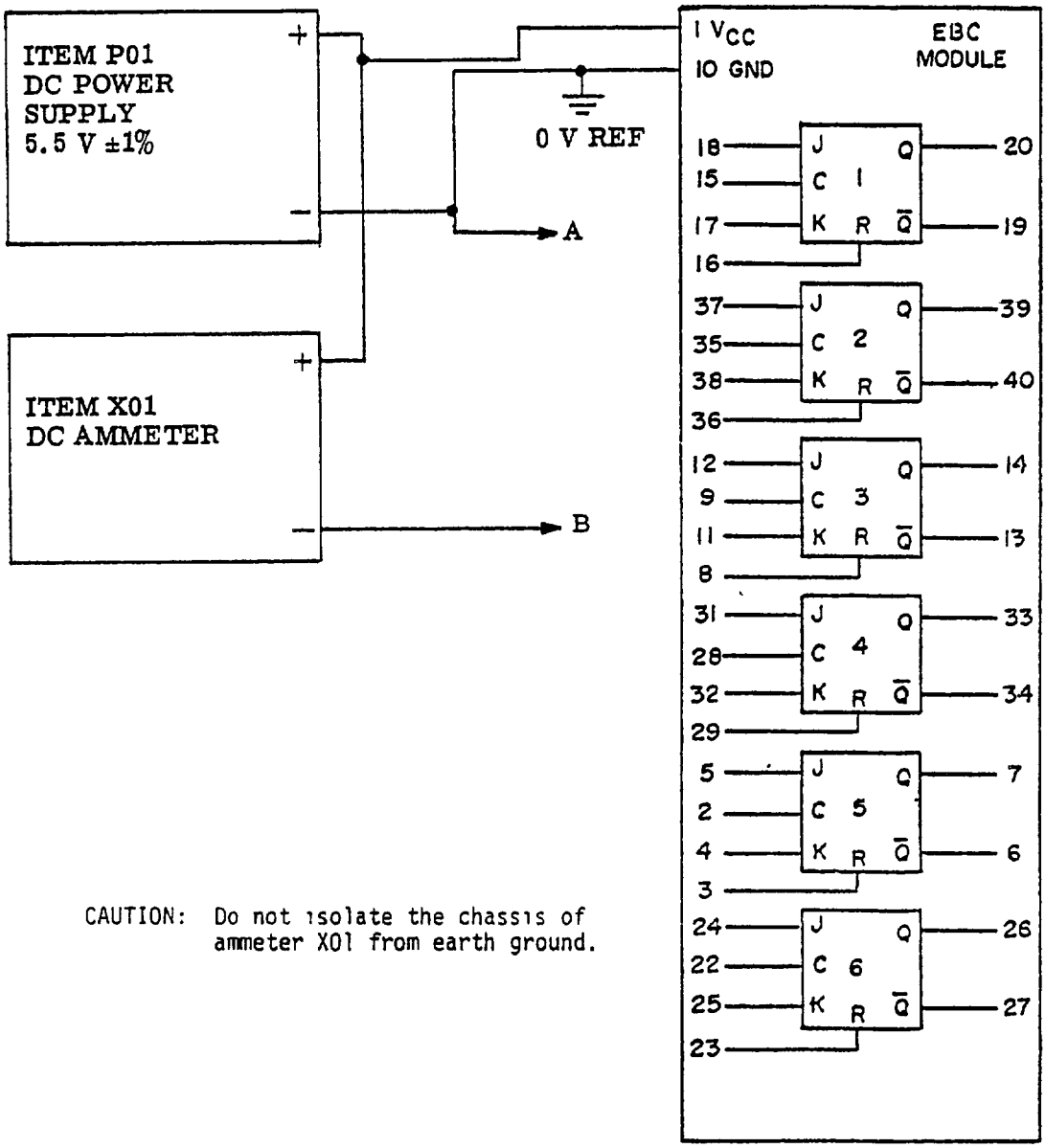
* FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	C1 Input breakdown current	17,18,16	15	0	200	µA	6
B2	C2	37,38,36	35				
B3	C3	11,12,8	9				
B4	C4	29,31,32	28				
B5	C5	3,4,5	2				
B6	C6	23,24,25	22				
B7	J1	15,16	18		100		
B8	J2	35,36	37				
B9	J3	8,9	12				
B10	J4	28,29	31				
B11	J5	2,3	5				
B12	J6	22,23	24				
B13	K1	15,16	17				
B14	K2	35,36	38				
B15	K3	8,9	11				
B16	K4	28,29	32				
B17	K5	2,3	4				
B18	K6	22,23	25				
B19	R1	15,18	16		200		
B20	R2	35,37	36				
B21	R3	9,12	8				
B22	R4	28,31	29				
B23	R5	2,5	3				
B24	R6	22,24	23				

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CAUTION: Do not isolate the chassis of ammeter X01 from earth ground.

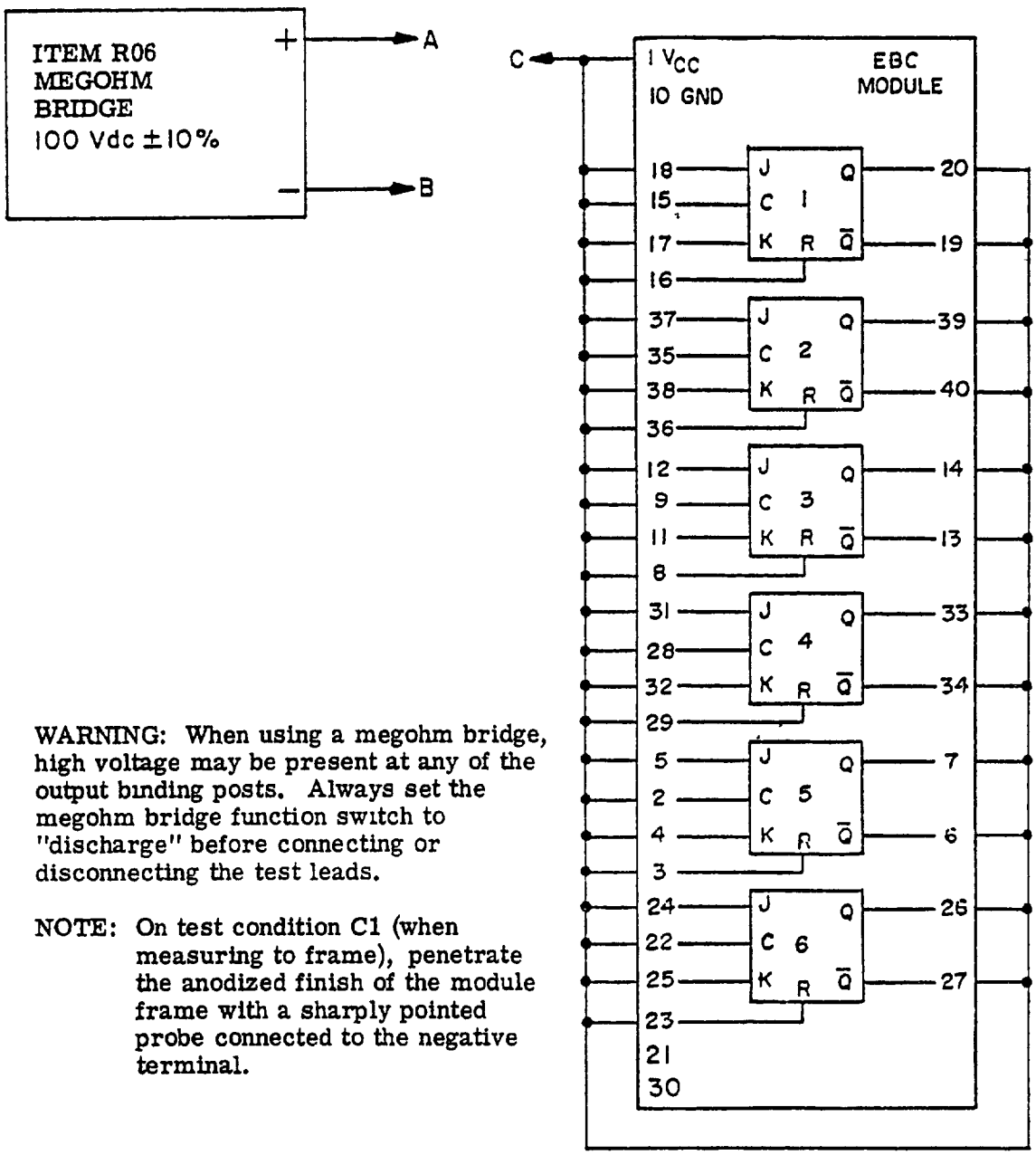
FIGURE 6. Input breakdown current test circuit for table IV.

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* TABLE V. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Isolation, frame to module pins	C,21,30	Frame	10	∞	MΩ	7
C2	Isolation, used to unused pins	C	21,30	10	∞	MΩ	7
C3	Isolation, unused to unused pins	21	30	10	∞	MΩ	7

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WARNING: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: On test condition C1 (when measuring to frame), penetrate the anodized finish of the module frame with a sharply pointed probe connected to the negative terminal.

* FIGURE 7. Isolation test circuit for table V.

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* TABLE VI. Input current (high-level and low-level) test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
D1	C1 Input current, high level	16,17,18	15	Open	Open	Open	0	-200	µA	8
D2	C2	36,37,38	35							
D3	C3	8,11,12	9							
D4	C4	29,31,32	28							
D5	C5	3,4,5	2							
D6	C6	23,24,25	22							
D7	J1	15,16	18					10		
D8	J2	35,36	37							
D9	J3	8,9	12							
D10	J4	28,29	31							
D11	J5	2,3	5							
D12	J6	22,23	24							
D13	K1	15,16	17							
D14	K2	35,36	38							
D15	K3	9,8	11							
D16	K4	28,29	32							
D17	K5	2,3	4							
D18	K6	22,23	25							
D19	R1	15,18	16					20		
D20	R2	35,37	36							
D21	R3	9,12	8							
D22	R4	28,31	29							
D23	R5	2,5	3							
D24	R6	22,24	23							

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* TABLE VI. Input current (high-level and low-level) test - Continued.

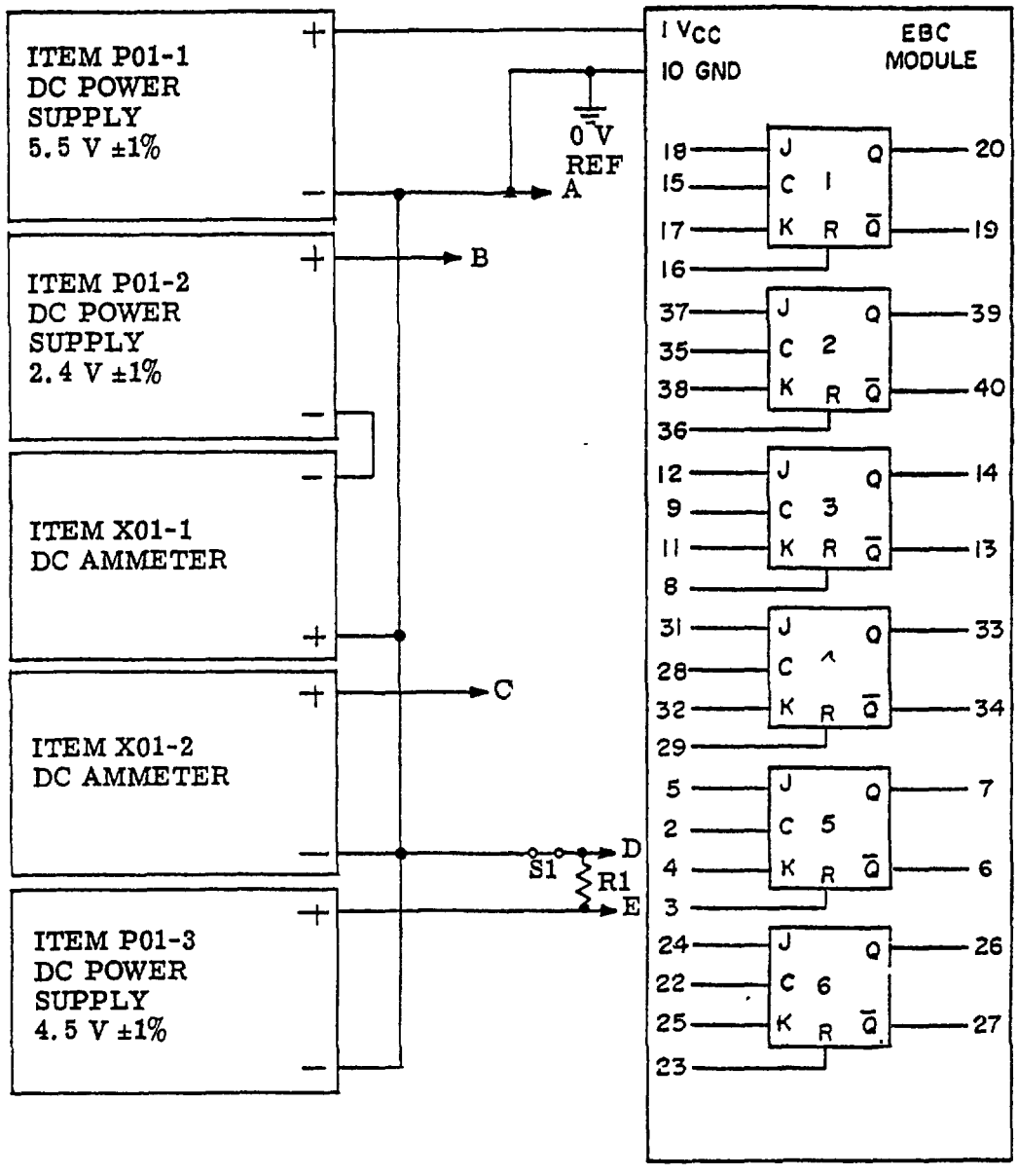
Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Connect E to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
E1	C1 Input current, low level	Open	Open	15	16	17,18	10	600	µA	8
E2	C2			35	36	37,38				
E3	C3			9	8	11,12				
E4	C4			28	29	31,32				
E5	C5			2	3	4,5				
E6	C6			22	23	24,25				
E7	J1			18	16	15,16		200		
E8	J2			37	36	35,36				
E9	J3			12	8	8,9				
E10	J4			31	29	28,29				
E11	J5			5	3	2,3				
E12	J6			24	23	22,23				
E13	K1			17	15	18,16				
E14	K2			38	35	37,36				
E15	K3			11	9	8,12				
E16	K4			32	28	31,29				
E17	K5			4	2	3,5				
E18	K6			25	22	24,23				
E19	R1			16	Open	15,18		400		
E20	R2			36		35,37				
E21	R3			8		9,12				
E22	R4			29		28,31				
E23	R5			3		2,5				
E24	R6			23		22,24				

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* TABLE VII. Output short circuit current test.

Test condition	Parameter	Connect: A to pin	B to pin	C to pin	D to pin	E to pin	Initial limits 25°±5°C ambient: Min	Max	Unit	Test figure
F1	Q1 Output short circuit current	17	Open	20	15	16,18	3	15	mA	8
F2	Q2	38		39	35	36,37				
F3	Q3	11		14	9	8,12				
F4	Q4	32		33	28	29,31				
F5	Q5	4		7	2	3,5				
F6	Q6	25		26	22	23,24				
F7	$\overline{Q}$1	16		19	Open	Open				
F8	$\overline{Q}$2	36		40						
F9	$\overline{Q}$3	8		13						
F10	$\overline{Q}$4	29		34						
F11	$\overline{Q}$5	3		6						
F12	$\overline{Q}$6	23		27						

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CAUTION: Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. R1 = 1.0 kΩ ±1%, 1/8 W.
2. S1 = SPST switch.
3. Since meters X01-1 and X01-2 are not used simultaneously, only one ammeter is required.
4. Since the power supplies P01-2 and P01-3 are not used simultaneously, only two item P01 supplies are required.
5. With point "B" disconnected, ammeter should indicate 0.0 current on the 1.0 µA range.
6. When performing output short circuit current test, only one output shall be shorted at a time, and the measurement taken within 5 seconds.
7. For test condition E, when using point D, make all circuit connections, open S1 for a minimum of 500 ns, close S1 for a minimum of 500 ns, then open it and make measurement.
8. For test condition F, when using point D, make all circuit connections, open S1 for a minimum of 500 ns, then close it and make measurement.

* FIGURE 8. Input current (high-level and low-level) and output short circuit current test circuit for tables VI and VII.

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TABLE VIII. Q and $\overline{Q}$ output voltage (high-level and low-level) test.

Test 1/ condition	Parameter	Connect							Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
G1	Q1 Output voltage, high level (J threshold high)	Open	17	18	16	15	20	19	2.4	4.5	V	9
H1	$\overline{Q}$1 Output voltage, high level (K threshold high)		18	17	Open	15	19	20				
G2	Q2 Output voltage, high level (J threshold high)		38	37	36	35	39	40				
H2	$\overline{Q}$2 Output voltage, high level (K threshold high)		37	38	Open	35	40	39				
G3	Q3 Output voltage, high level (J threshold high)		11	12	8	9	14	13				
H3	$\overline{Q}$3 Output voltage, high level (K threshold high)		12	11	Open	9	13	14				
G4	Q4 Output voltage, high level (J threshold high)		32	31	29	28	33	34				
H4	$\overline{Q}$4 Output voltage, high level (K threshold high)		31	32	Open	28	34	33				
G5	Q5 Output voltage, high level (J threshold high)		4	5	3	2	7	6				
H5	$\overline{Q}$5 Output voltage, high level (K threshold high)		5	4	Open	2	6	7				
G6	Q6 Output voltage, high level (J threshold high)		25	24	23	22	26	27				
H6	$\overline{Q}$6 Output voltage, high level (K threshold high)		24	25	Open	22	27	26				
J1	$\overline{Q}$1 Output voltage, high level (Reset threshold low)	18	17	16		15	19	20				
J2	$\overline{Q}$2	37	38	36		35	40	39				
J3	$\overline{Q}$3	12	11	8		9	13	14				
J4	$\overline{Q}$4	31	32	29		28	34	33				
J5	$\overline{Q}$5	5	4	3		2	6	7				

See footnote at end of table.

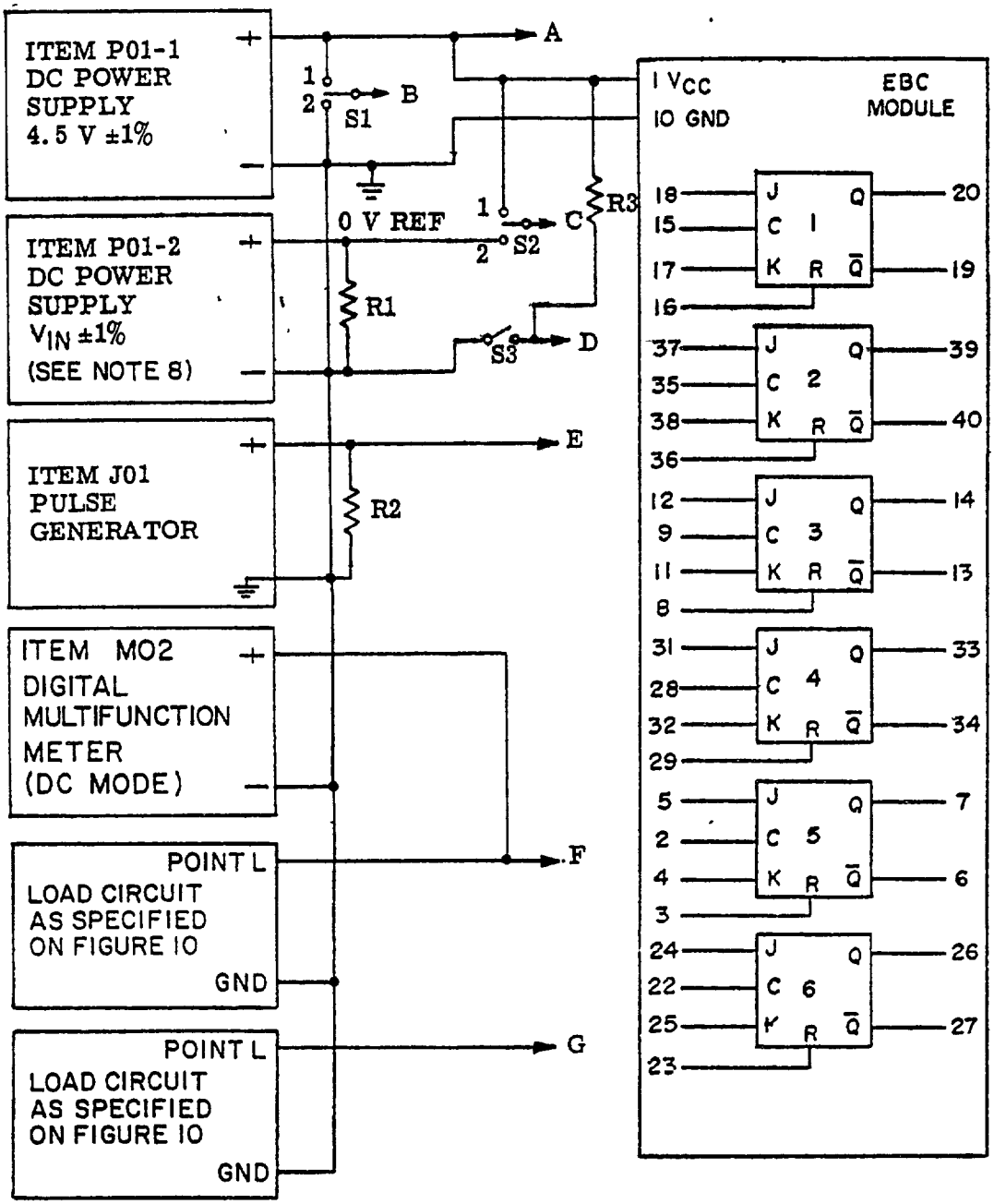
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TABLE VIII. Q and $\overline{Q}$ output voltage (high-level and low-level) test - Continued.

Test 1/ condition	Parameter	Connect							Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
J6	$\overline{Q}$6 Output voltage, high level (Reset threshold low)	24	25	23	Open	22	27	26	2.4	4.5	V	9
J7	Q1 Output voltage, high level (Reset threshold high)	18	17	16		15	20	19				
J8	Q2	37	38	36		35	39	40				
J9	Q3	12	11	8		9	14	13				
J10	Q4	31	32	29		28	33	34				
J11	Q5	5	4	3		2	7	6				
J12	Q6	24	25	23		22	26	27				
K1	Q1 Output voltage, low level (J threshold low)	Open	17	18	16	15	20	19	0.0	0.3		
K2	Q2		38	37	36	35	39	40				
K3	Q3		11	12	8	9	14	13				
K4	Q4		32	31	29	28	33	34				
K5	Q5		4	5	3	2	7	6				
K6	Q6		25	24	23	22	26	27				
K7	$\overline{Q}$1 Output voltage, low level (K threshold low)		Open	17	16	15	19	20				
K8	$\overline{Q}$2			38	36	35	40	39				
K9	$\overline{Q}$3			11	8	9	13	14				
K10	$\overline{Q}$4			32	29	28	34	33				
K11	$\overline{Q}$5			4	3	2	6	7				
K12	$\overline{Q}$6			25	23	22	27	26				

1/ For test conditions G and H see note 5, figure 9.

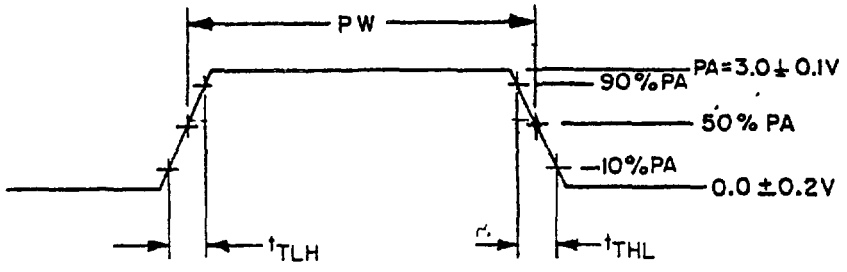
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* FIGURE 9. Q and $\overline{Q}$ output voltage (high-level and low-level) test circuit for table VIII.

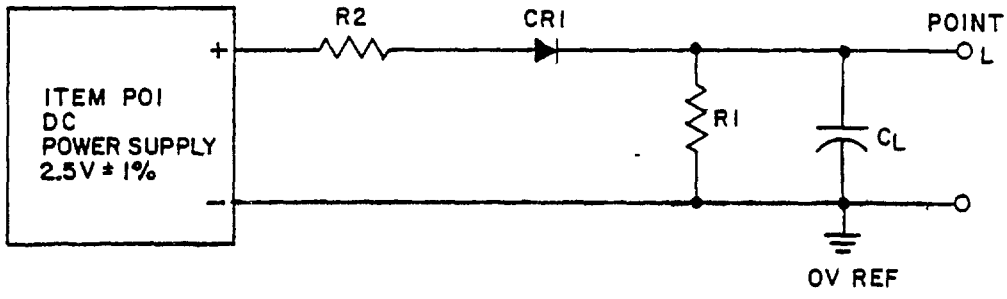
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NOTES:
1. R1=R2 = 50Ω ±5%.
2. R3 = 2.75 kΩ ±1%, 1/8 W.
3. S1, S2 = SPDT switch.
4. S3 = SPST switch.
5. Test conditions G and H are performed in sequence. Perform test condition G1. Without disconnecting the power or point C, setup for test condition H1. Perform test condition H1. Repeat this sequence for G2 and H2 through G6 and H6 (the G test conditions set Q at the high state which is necessary before the H test conditions can be performed).
6. Switching sequence for test conditions G, H, J, and K7-K12.
   (a) Initially set S1 and S2 in position 2 and S3 open.
   (b) Close S3 for a minimum of 500 ns, then open it.
   (c) Apply one pulse with item J01 and make measurement.
7. Switching sequence for test conditions K1-K6.
   (a) Initially set S1 in position 2 and S2 in position 1, S3 open.
   (b) Apply one pulse with item J01.
   (c) Place S1 in position 1 and S2 in position 2.
   (d) Apply one pulse with item J01 and make measurement.
8. For test conditions G, H, and J7-J12, V_{IN} = 2.0 V.
   For test conditions J1-J6 and K, V_{IN} = 0.7 V.
9. Item J01 setup:
   Single cycle mode.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   PW = 500 ±50 ns.



* FIGURE 9. Q and $\overline{Q}$ output voltage (high-level and low-level) test circuit for table VIII.- Continued.

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NOTES:
1. CR1 = 1N916, or equivalent.
2. R1 = 23.7 KΩ ±1%, 1/8 W with a current flow of 100 µA out of point L when point "L" is held at 2.4 V.
3. R2 = Select value for a current flow of 2.0 mA ±5% out of point L when point "L" is held at 0.3 V.
4. For test conditions G1-G6, H1-H6, J1-J12, K1-K12, M1-M12, N1-N12, P1-P12, Q1-Q12, R1-R12, S1-S12, T1-T12, U1-U6, and V1-V6, and C_L = 40 ±5 pF (including probe and parasitic capacitance).
5. For test conditions N13-N24, P13-P24, R13-R24, S13-S24, and T13-T24, C_L = 140 ±5 pF (including probe and parasitic capacitance).
6. For test conditions W1-W12 and X1-X12, C_L = 50 ±5 pF (including probe and parasitic capacitance).
7. C_L is not required for dc electrical test conditions.
8. The maximum lead length from point "L" to the appropriate module pin shall be 3 inches (76.2 mm).
9. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV.

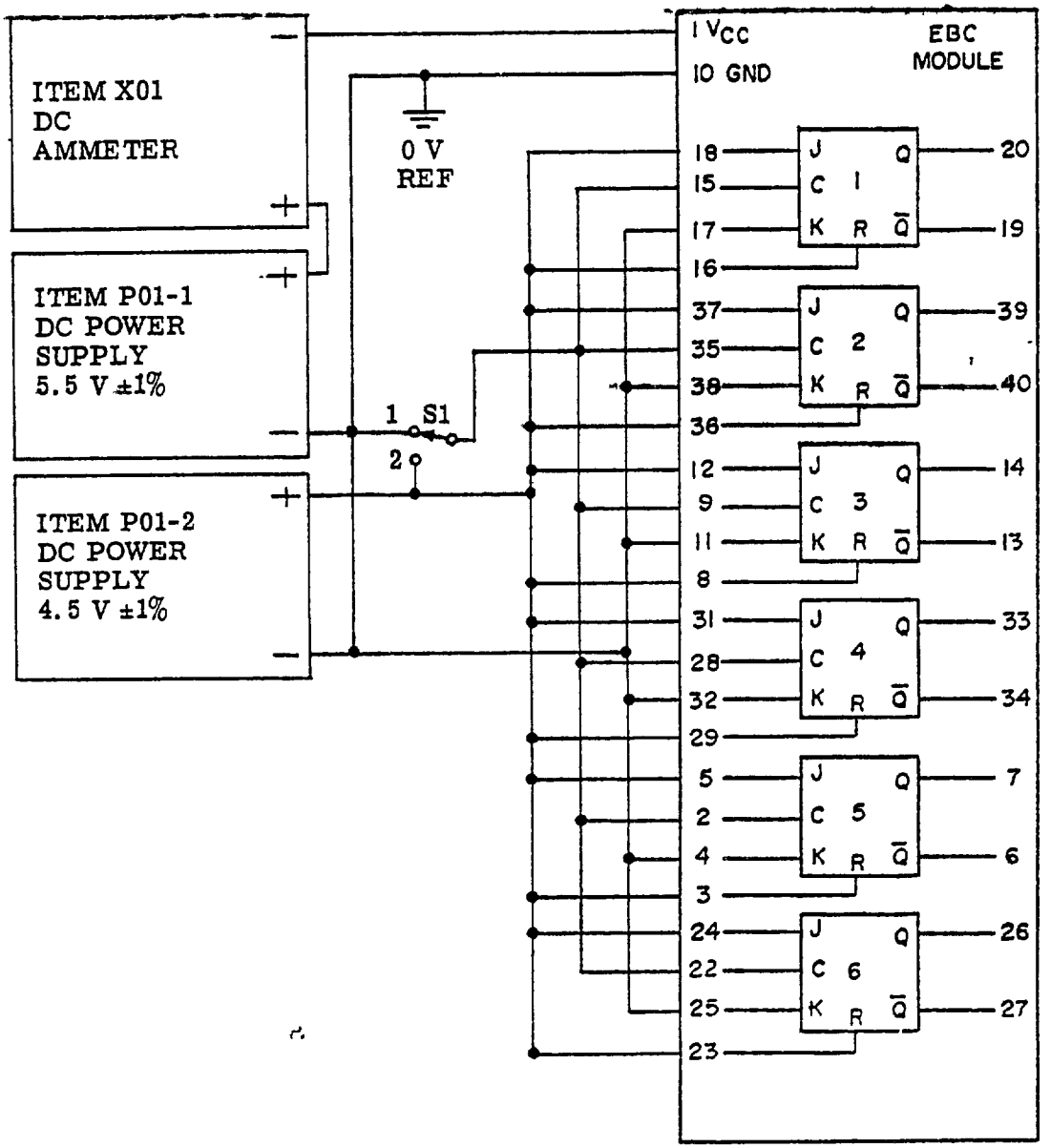
* FIGURE 10. Logic load circuit.

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TABLE IX. Power supply current test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
L	Power supply current (Q outputs high level)	2.0	8.6	mA	11

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CAUTION: Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. S1 = SPDT switch.
2. After making all circuit connections, initially set S1 in position 1, place S1 in position 2 for a minimum of 500 ns, then place S1 back in position 1 and make measurement.

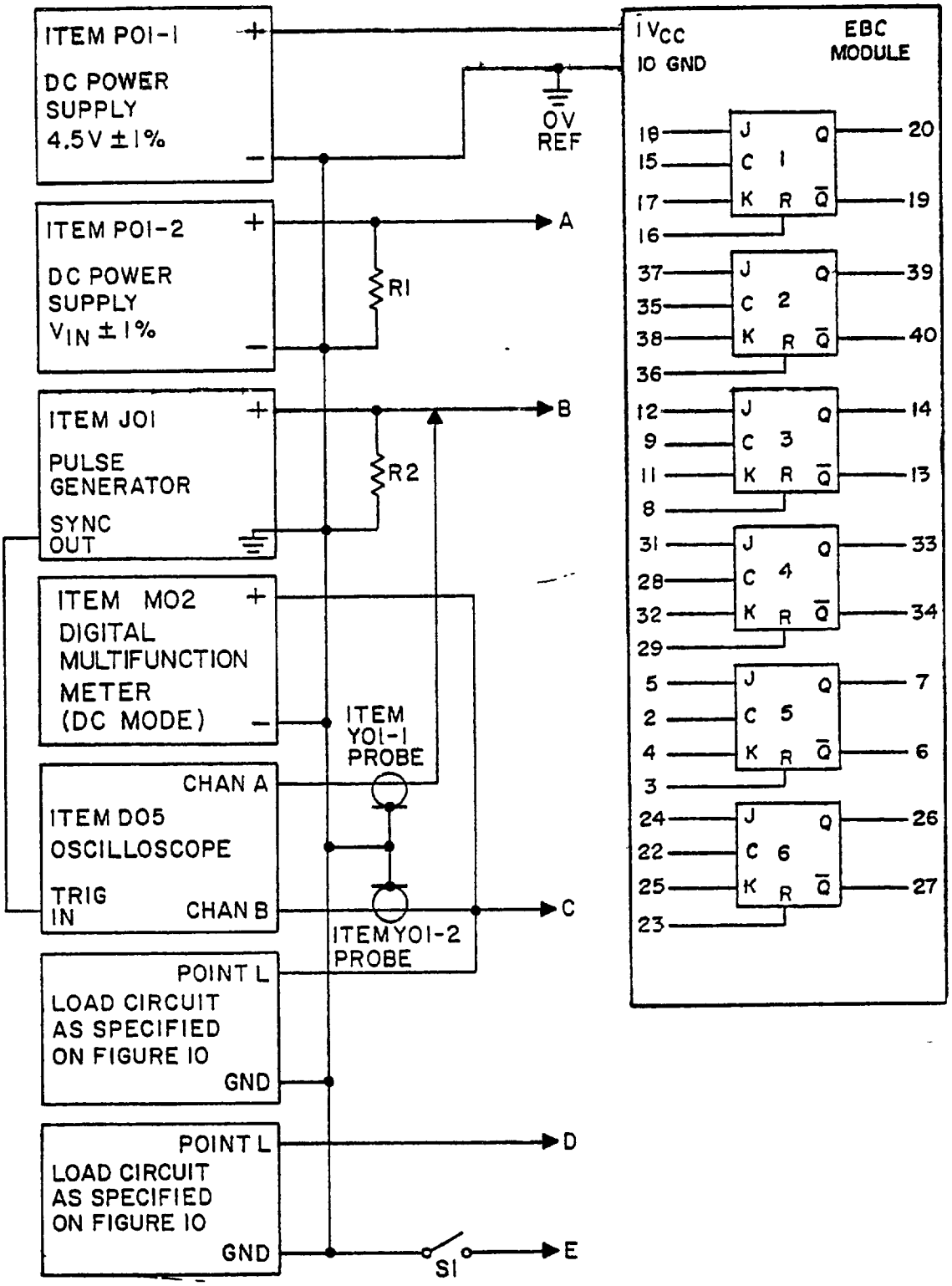
FIGURE 11. Power supply current test circuit for table IX.

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TABLE X. JK operation (J and K threshold high and low; maximum clock frequency and minimum pulse width) test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
M1	J1K1 Operation (J and K threshold high)	17,18	15	20	19	Open	760	840	ns	12,13
M2	J2K2	37,38	35	39	40					
M3	J3K3	11,12	9	14	13					
M4	J4K4	31,32	28	33	34					
M5	J5K5	4,5	2	7	6					
M6	J6K6	24,25	22	26	27					
M7	J1K1 Operation (J and K threshold low)	17,18	15	20	19	16	0.0	0.3	V	
M8	J2K2	37,38	35	39	40	36				
M9	J3K3	11,12	9	14	13	8				
M10	J4K4	31,32	28	33	34	29				
M11	J5K5	4,5	2	7	6	3				
M12	J6K6	24,25	22	26	27	23				

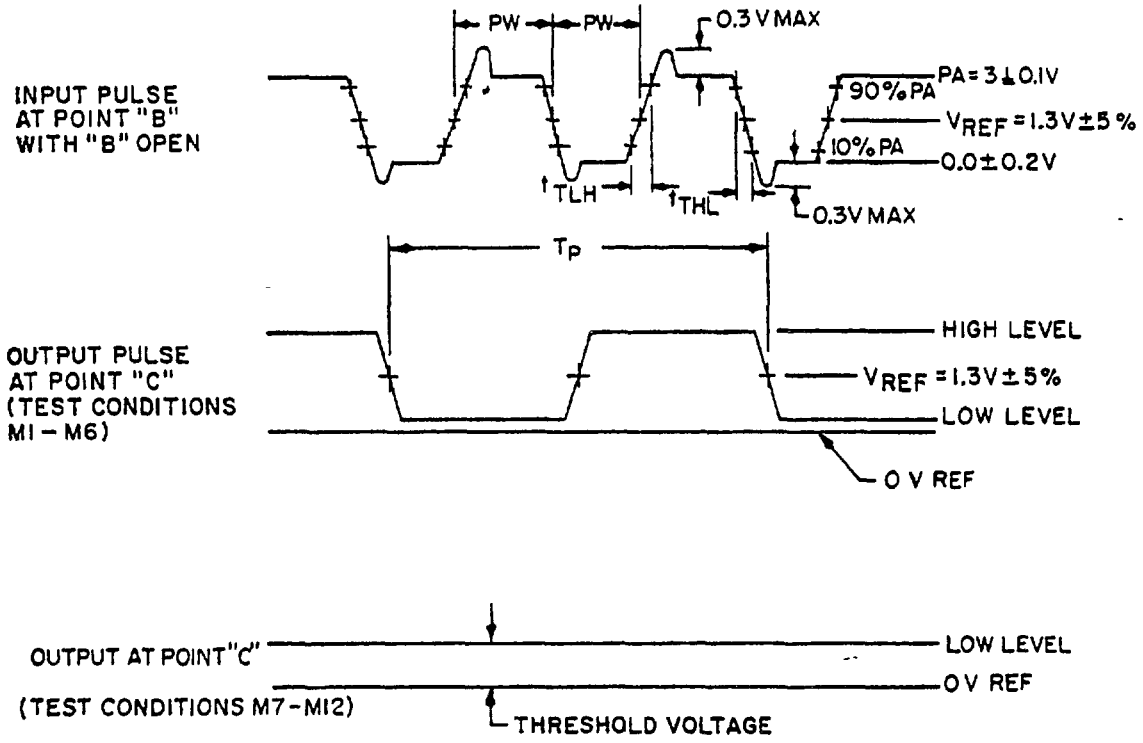
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NOTES:
1. R1 = R2 = 50Ω ±5%, 1 W.
2. S1 = SPST switch.
3. Switching sequence for test condition M7-M12. Close switch S1 for a minimum of 500 ns, then open it and make measurement.
4. For test condition M7-M12, read output voltage level with item M02.
5. For test condition M1-M6, V_{IN} = 2.0 V and for test condition M7-M12, V_{IN} = 0.7 V.

* FIGURE 12. J K operation (J and K threshold high and low; maximum clock frequency; minimum pulse width) test circuit for table X.

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NOTES:
1. Input pulse parameters:
   PW = 200 ±5 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. All voltage levels are referenced to ground.

* FIGURE 13. J K operation (J and K threshold high and low; maximum clock frequency; minimum pulse width) test circuit wavefroms for table X.

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* TABLE XI. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
N1	Q1 Transition time (t_{TLH}), C_L = 40 pF	15,17,18	20	19	5	60	ns	14,15
N2	Q2	35,37,38	39	40				
N3	Q3	9,11,12	14	13				
N4	Q4	28,31,32	33	34				
N5	Q5	2,4,5	7	6				
N6	Q6	22,24,25	26	27				
N7	$\overline{Q}$1	15,17,18	19	20				
N8	$\overline{Q}$2	35,37,38	40	39				
N9	$\overline{Q}$3	9,11,12	13	14				
N10	$\overline{Q}$4	28,31,32	34	33				
N11	$\overline{Q}$5	2,4,5	6	7				
N12	$\overline{Q}$6	22,24,25	27	26				
N13	Q1 Transition time (t_{TLH}), C_L = 140 pF	15,17,18	20	19	10	100		
N14	Q2	35,37,38	39	40				
N15	Q3	9,11,12	14	13				
N16	Q4	28,31,32	33	34				
N17	Q5	2,4,5	7	6				
N18	Q6	22,24,25	26	27				
N19	$\overline{Q}$1	15,17,18	19	20				
N20	$\overline{Q}$2	35,37,38	40	39				
N21	$\overline{Q}$3	9,11,12	13	14				
N22	$\overline{Q}$4	28,31,32	34	33				
N23	$\overline{Q}$5	2,4,5	6	7				
N24	$\overline{Q}$6	22,24,25	27	26				

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* TABLE XI. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P1	Q1 Transition time (t_{THL}), C_L = 40 pF	15, 17,18	20	19	2	30	ns	14,15
P2	Q2	35,37,38	39	40				
P3	Q3	9,11,12	14	13				
P4	Q4	28,31,32	33	34				
P5	Q5	2,4,5	7	6				
P6	Q6	22,24,25	26	27				
P7	$\overline{Q}$1	15,17,18	19	20				
P8	$\overline{Q}$2	35,37,38	40	39				
P9	$\overline{Q}$3	9,11,12	13	14				
P10	$\overline{Q}$4	28,31,32	34	33				
P11	$\overline{Q}$5	2,4,5	6	7				
P12	$\overline{Q}$6	22,24,25	27	26				
P13	Q1 Transition time (t_{THL}), C_L = 140 pF	15,17,18	20	19	5	60		
P14	Q2	35,37,38	39	40				
P15	Q3	9,11,12	14	13				
P16	Q4	28,31,32	33	34				
P17	Q5	2,4,5	7	6				
P18	Q6	22,24,25	26	27				
P19	$\overline{Q}$1	15,17,18	19	20				
P20	$\overline{Q}$2	35,37,38	40	39				
P21	$\overline{Q}$3	9,11,12	13	14				
P22	$\overline{Q}$4	28,31,32	34	33				
P23	$\overline{Q}$5	2,4,5	6	7				
P24	$\overline{Q}$6	22,24,25	27	26				

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* TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test.

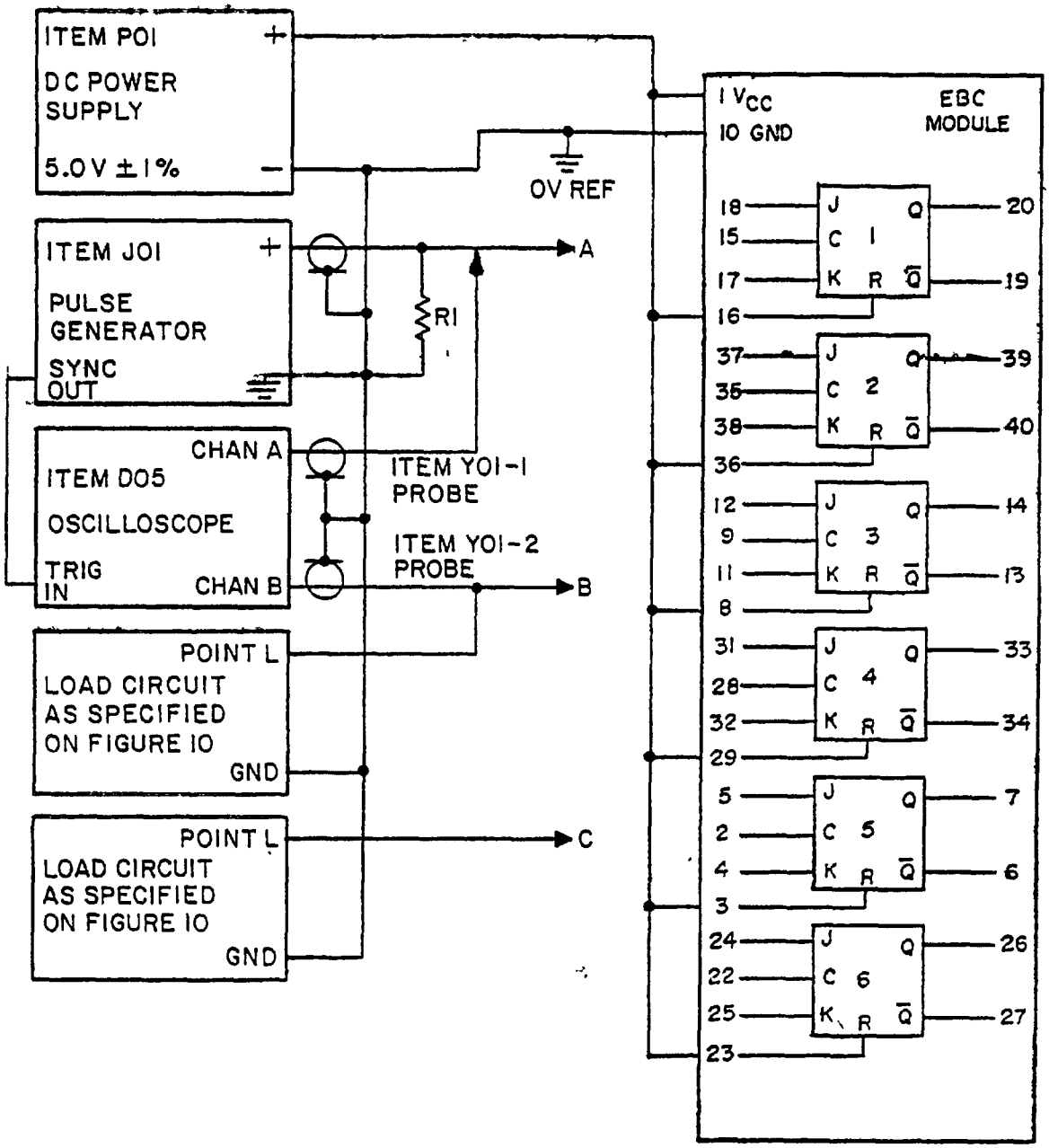
Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
Q1	C1 to Q1 Propagation delay time (t_{PLH}), C_L = 40 pF	15,17,18	20	19	10	75	ns	14,15
Q2	C2 to Q2	35,37,38	39	40				
Q3	C3 to Q3	9,11,12	14	13				
Q4	C4 to Q4	28,31,32	33	34				
Q5	C5 to Q5	2,4,5	7	6				
Q6	C6 to Q6	22,24,25	26	27				
Q7	C1 to $\overline{Q}$1	15,17,18	19	20				
Q8	C2 to $\overline{Q}$2	35,37,38	40	39				
Q9	C3 to $\overline{Q}$3	9,11,12	13	14				
Q10	C4 to $\overline{Q}$4	28,31,32	34	33				
Q11	C5 to $\overline{Q}$5	2,4,5	6	7				
Q12	C6 to $\overline{Q}$6	22,24,25	27	26				
Q13	C1 to Q1 Propagation delay time (t_{PLH}), C_L = 140 pF	15,17,18	20	19		105		
Q14	C2 to Q2	35,37,38	39	40				
Q15	C3 to Q3	9,11,12	14	13				
Q16	C4 to Q4	28,31,32	33	34				
Q17	C5 to Q5	2, 4, 5	7	6				
Q18	C6 to Q6	22,24,25	26	27				
Q19	C1 to $\overline{Q}$1	15,17,18	19	20				
Q20	C2 to $\overline{Q}$2	35,37,38	40	39				
Q21	C3 to $\overline{Q}$3	9,11,12	13	14				
Q22	C4 to $\overline{Q}$4	28,31,32	34	33				
Q23	C5 to $\overline{Q}$5	2,4,5	6	7				
Q24	C6 to $\overline{Q}$6	22,24,25	27	26				

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* TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
R1	C1 to Q1 Propagation delay time (t_{PHL}), C_L = 40 pF	15,17,18	20	19	10	150	ns	14,15
R2	C2 to Q2	35,37,38	39	40				
R3	C3 to Q3	9,11,12	14	13				
R4	C4 to Q4	28,31,32	33	34				
R5	C5 to Q5	2,4,5	7	6				
R6	C6 to Q6	22,24,25	26	27				
R7	C1 to $\overline{Q}$1	15,17,18	19	20				
R8	C2 to $\overline{Q}$2	35,37,38	40	39				
R9	C3 to $\overline{Q}$3	9,11,12	13	14				
R10	C4 to $\overline{Q}$4	28,31,32	34	33				
R11	C5 to $\overline{Q}$5	2,4,5	6	7				
R12	C6 to $\overline{Q}$6	22,24,25	27	26				
R13	C1 to Q1 Propagation delay time (t_{PHL}), C_L = 140 pF	15,17,18	20	19		180		
R14	C2 to Q2	35,37,38	39	40				
R15	C3 to Q3	9,11,12	14	13				
R16	C4 to Q4	28,31,32	33	34				
R17	C5 to Q5	2,4,5	7	6				
R18	C6 to Q6	22,24,25	26	27				
R19	C1 to $\overline{Q}$1	15,17,18	19	20				
R20	C2 to $\overline{Q}$2	35,37,38	40	39				
R21	C3 to $\overline{Q}$3	9,11,12	13	14				
R22	C4 to $\overline{Q}$4	28,31,32	34	33				
R23	C5 to $\overline{Q}$5	2,4,5	6	7				
R24	C6 to $\overline{Q}$6	22,24,25	27	26				

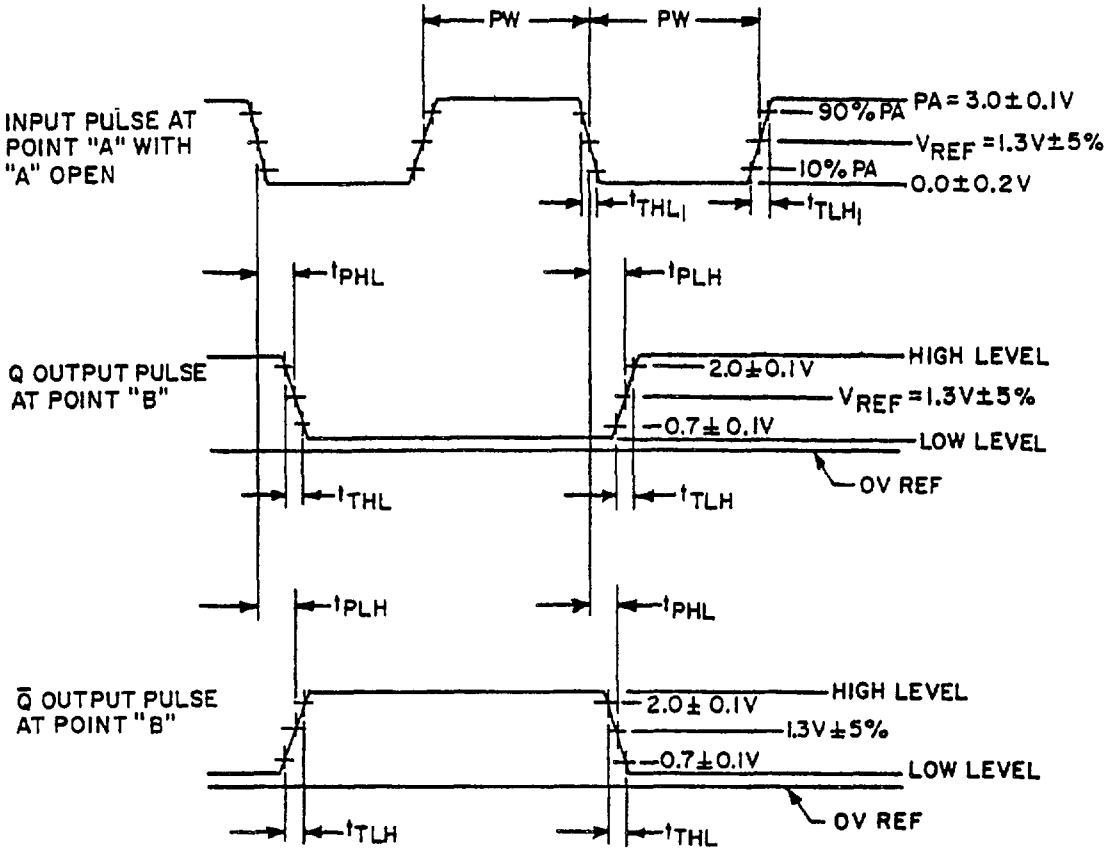
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NOTE: R1 = 50Ω ±5%, 1 W.

* FIGURE 14. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XI and XII.

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NOTES:
1. Input pulse parameters:
   PW = 500 ±50 ns.
   $t_{TLH_1} = t_{THL_1}$ = 10 ±2 ns.
2. Since the J, K and clock inputs are connected together, the requirements of an input setup time of ≥ clock pulse width and an input hold time of 0 ns is checked.
3. Before making the output pulse characterisitc measurements, the output levels shall be verified to be 0.3 V maximum for the low level and 2.4 minimum for the high level.
4. All voltage levels are referenced to ground.

* FIGURE 15. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XI and XII.

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* TABLE XIII. Propagation delay time (t_{PHL} and t_{PLH}) (minimum reset pulse width check) test.

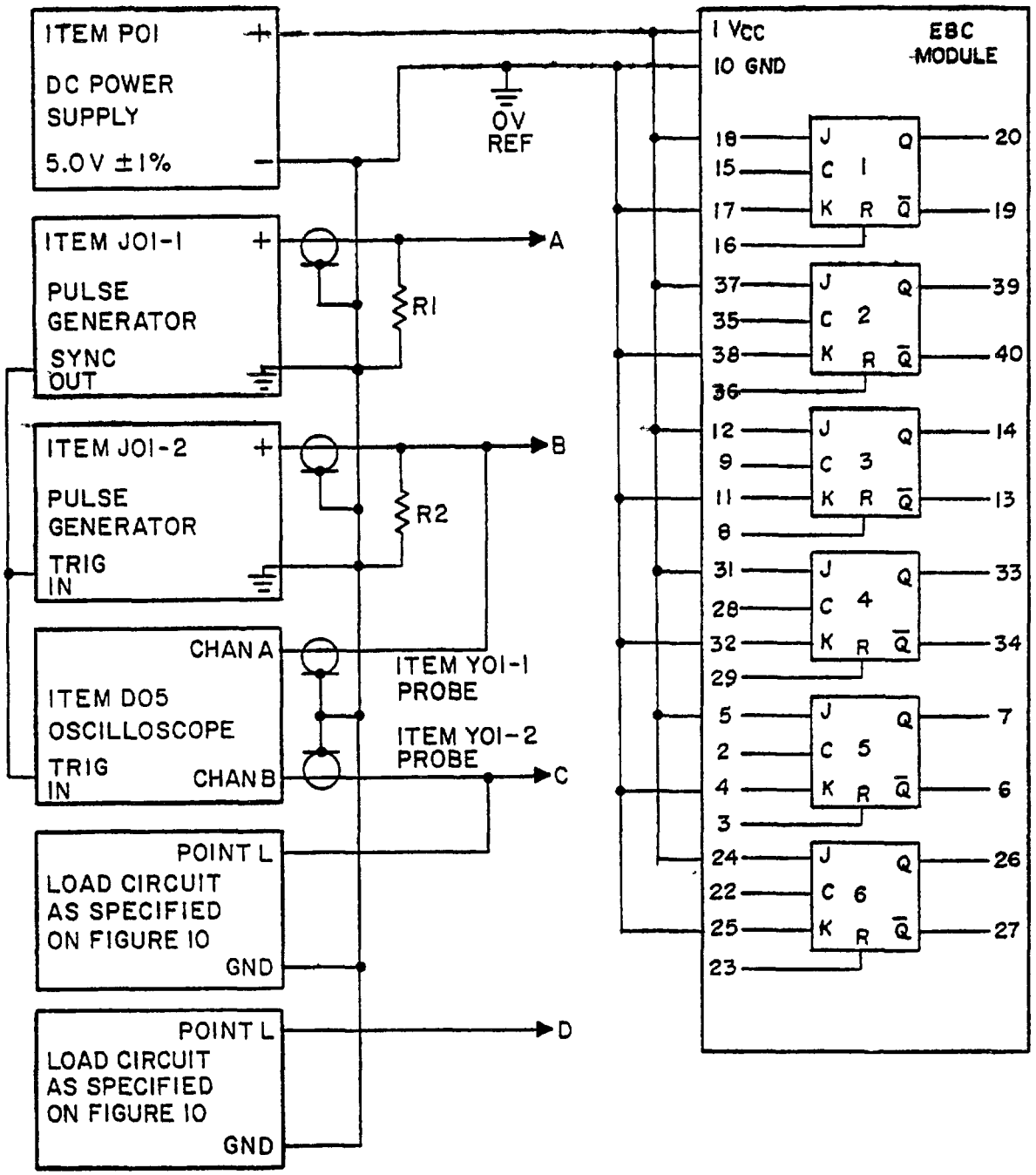
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	R1 to Q1 Propagation delay time (t_{PHL}), (V_{IN} clock = 0 V), C_L = 40 pF	15	16	20	19	10	200	ns	16,17
S2	R2 to Q2	35	36	39	40				
S3	R3 to Q3	9	8	14	13				
S4	R4 to Q4	28	29	33	34				
S5	R5 to Q5	2	3	7	6				
S6	R6 to Q6	22	23	26	27				
S7	R1 to $\overline{Q1}$ Propagation delay time (t_{PLH}), (V_{IN} clock = 2.4 V), C_L = 40 pF	15	16	19	20		75		
S8	R2 to $\overline{Q2}$	35	36	40	39				
S9	R3 to $\overline{Q3}$	9	8	13	14				
S10	R4 to $\overline{Q4}$	28	29	34	33				
S11	R5 to $\overline{Q5}$	2	3	6	7				
S12	R6 to $\overline{Q6}$	22	23	27	26				
S13	R1 to Q1 Propagation delay time (t_{PHL}), (V_{IN} clock = 0 V), C_L = 140 pF	15	16	20	19		230		
S14	R2 to Q2	35	36	39	40				
S15	R3 to Q3	9	8	14	13				
S16	R4 to Q4	28	29	33	34				
S17	R5 to Q5	2	3	7	6				
S18	R6 to Q6	22	23	26	27				
S19	R1 to $\overline{Q1}$ Propagation delay time (t_{PLH}), (V_{IN} clock = 2.4 V), C_L = 140 pF	15	16	19	20		105		
S20	R2 to $\overline{Q2}$	35	36	40	39				
S21	R3 to $\overline{Q3}$	9	8	13	14				
S22	R4 to $\overline{Q4}$	28	29	34	33				
S23	R5 to $\overline{Q5}$	2	3	6	7				
S24	R6 to $\overline{Q6}$	22	23	27	26				

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* TABLE XIII. Propagation delay time (t_{PHL} and t_{PLH}) (minimum reset pulse width check) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T1	R1 to $\overline{Q}1$ Propagation delay time (t_{PLH}), (V_{IN} clock = 0 V), C_L = 40 pF	15	16	19	20	10	75	ns	16,17
T2	R2 to $\overline{Q}2$	35	36	40	39				
T3	R3 to $\overline{Q}3$	9	8	13	14				
T4	R4 to $\overline{Q}4$	28	29	34	33				
T5	R5 to $\overline{Q}5$	2	3	6	7				
T6	R6 to $\overline{Q}6$	22	23	27	26				
T7	R1 to Q1 Propagation delay time (t_{PHL}), (V_{IN} clock = 2.4 V), C_L = 40 pF	15	16	20	19		150		
T8	R2 to Q2	35	36	39	40				
T9	R3 to Q3	9	8	14	13				
T10	R4 to Q4	28	29	33	34				
T11	R5 to Q5	2	3	7	6				
T12	R6 to Q6	22	23	26	27				
T13	R1 to $\overline{Q}1$ Propagation delay time (t_{PLH}), (V_{IN} clock = 0 V), C_L = 140 pF	15	16	19	20		105		
T14	R2 to $\overline{Q}2$	35	36	40	39				
T15	R3 to $\overline{Q}3$	9	8	13	14				
T16	R4 to $\overline{Q}4$	28	29	34	33				
T17	R5 to $\overline{Q}5$	2	3	6	7				
T18	R6 to $\overline{Q}6$	22	23	27	26				
T19	R1 to Q1 Propagation delay time (t_{PHL}), (V_{IN} clock = 2.4 V), C_L = 140 pF	15	16	20	19		180		
T20	R2 to Q2	35	36	39	40				
T21	R3 to Q3	9	8	14	13				
T22	R4 to Q4	28	29	33	34				
T23	R5 to Q5	2	3	7	6				
T24	R6 to Q6	22	23	26	27				

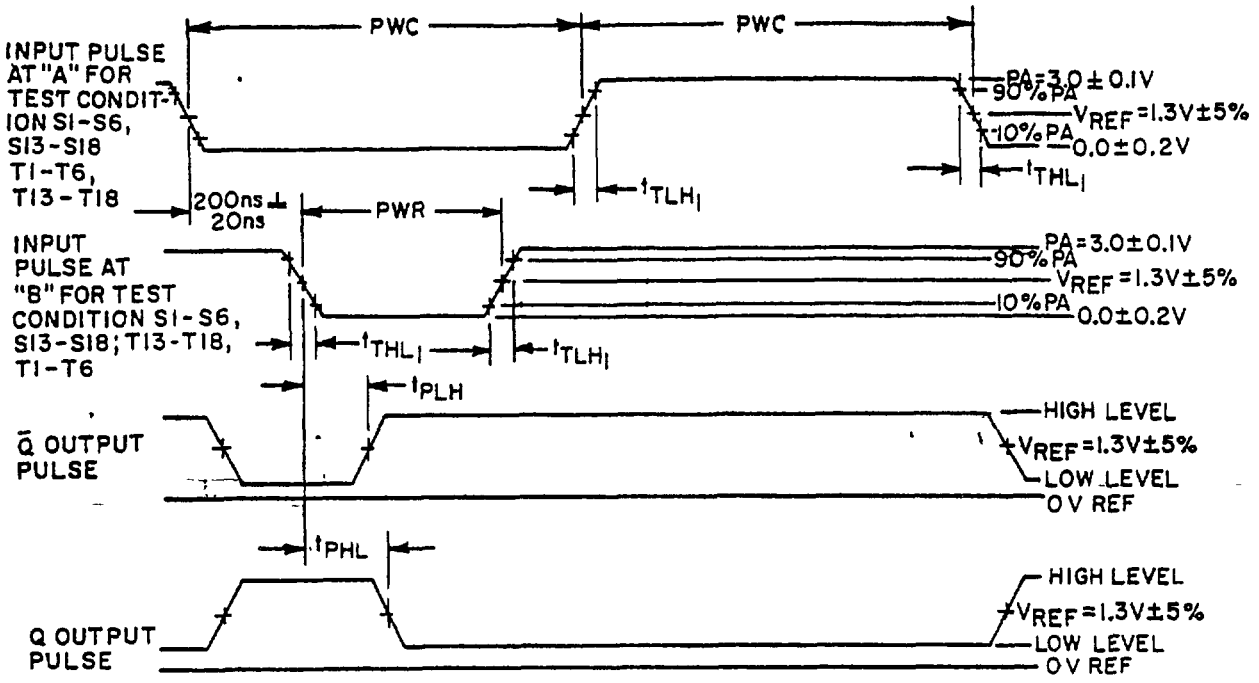
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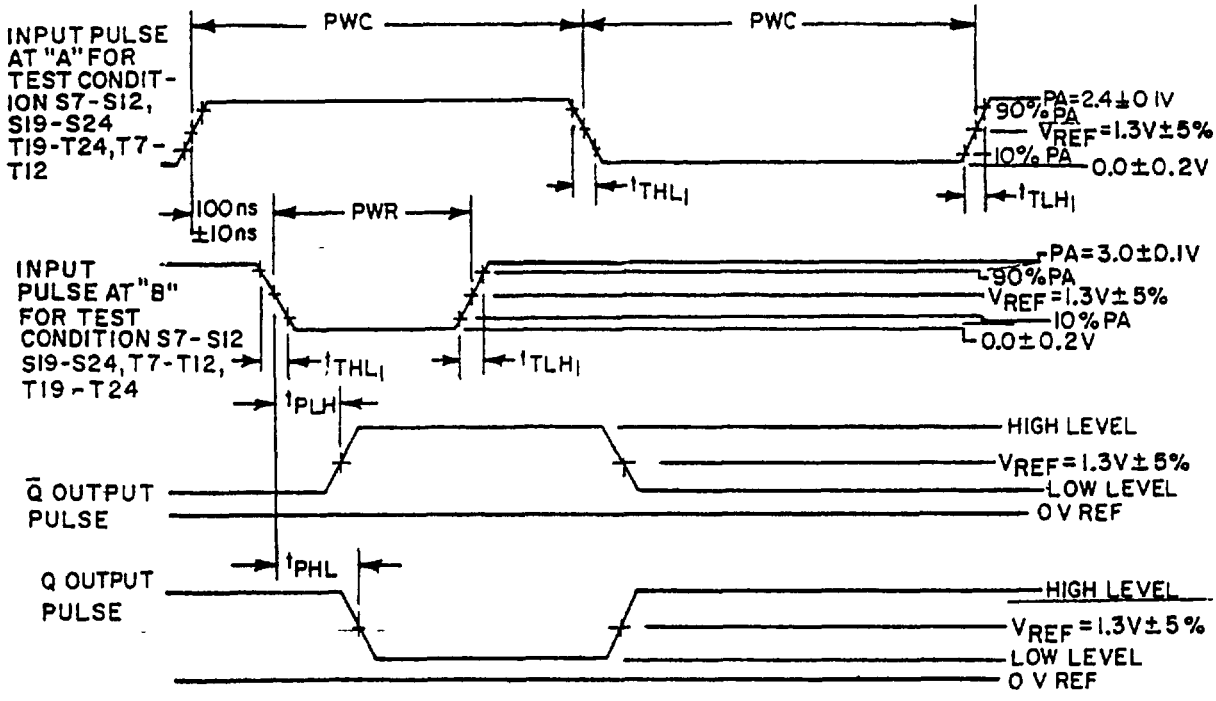


NOTE: R1 = R2 = 50Ω ±5%, 1 W.

* FIGURE 16. Propagation delay time (t_{PHL} and t_{PLH}) (minimum reset pulse width check) test circuit for table XIII.

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NOTES:
1. Input pulse parameters:
   Clock pulse width (PWC) = 400 ±40 ns.
   Reset pulse width (PWR) = 100 ±10 ns.
   $t_{TLH_1} = t_{THL_1} = 10 \pm 2$ ns.
2. These tests verify minimum reset pulse width of 100 ns.
3. All voltage levels are referenced to ground.

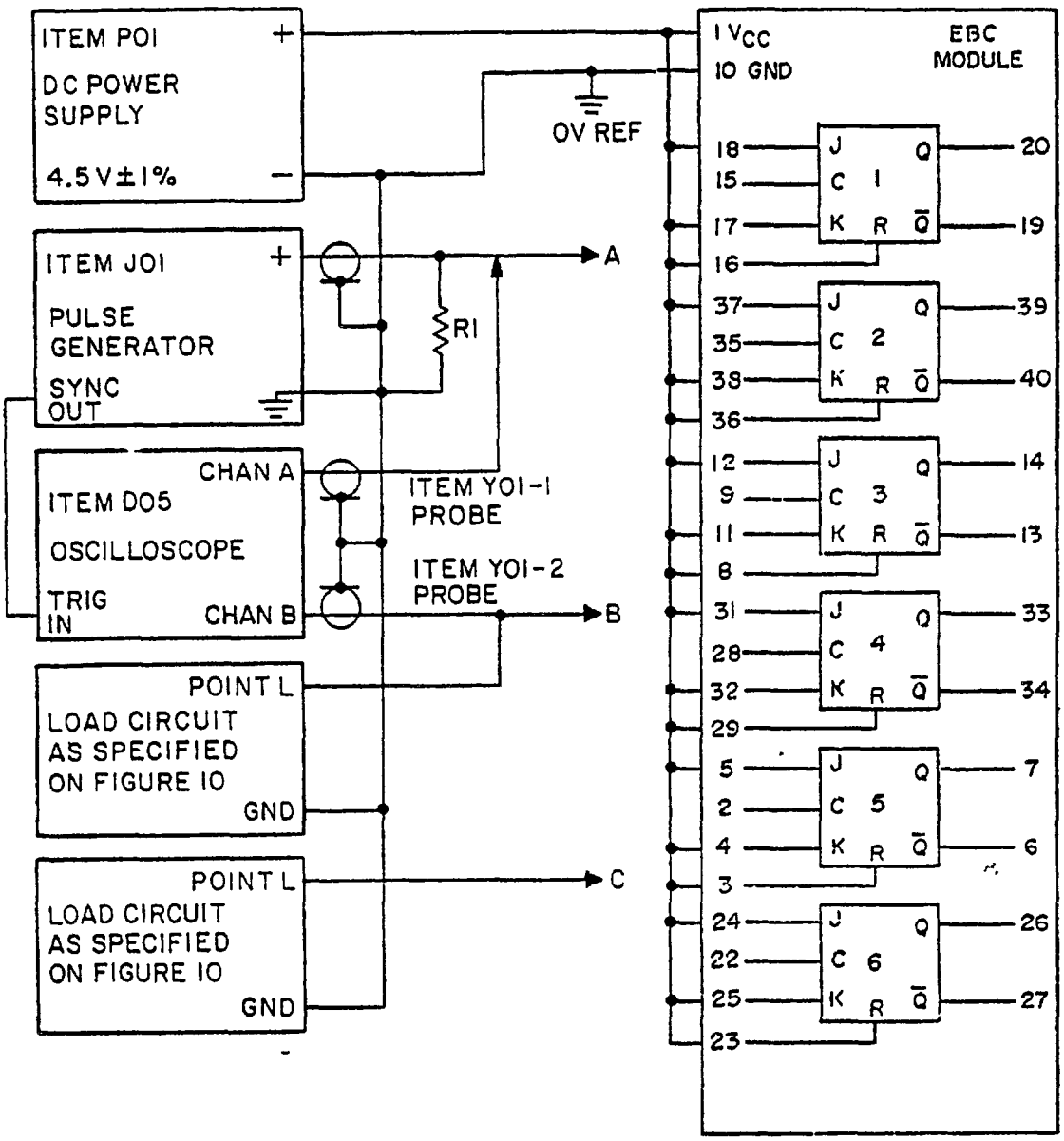
* FIGURE 17. Propagation delay time (t_{PHL} and t_{PLH}) (minimum reset pulse width check) test circuit waveforms for table XIII.

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TABLE XIV. Maximum clock fall time and threshold clock input test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin.	B to pin	C to pin	Min	Max		
U1	C1 Maximum clock fall time	15	20	19	5.7	6.3	µs	18,19
U2	C2	35	39	40				
U3	C3	9	14	13				
U4	C4	28	33	34				
U5	C5	2	7	6				
U6	C6	22	26	27				
V1	C1 Threshold clock input	15	19	20	0.7	0.9		
V2	C2	35	40	39				
V3	C3	9	13	14				
V4	C4	28	34	33				
V5	C5	2	6	7				
V6	C6	22	27	26				

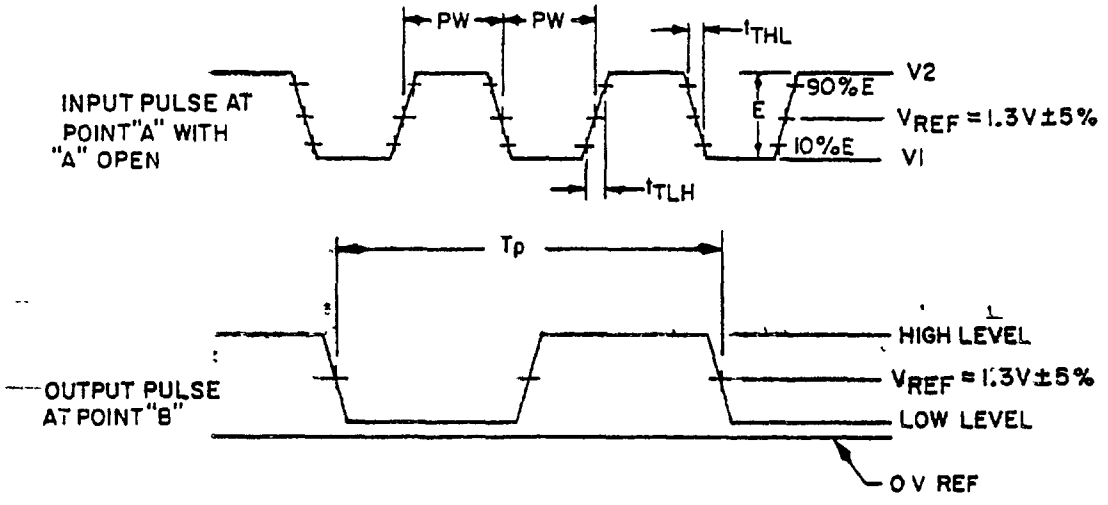
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NOTE: R1 = 50Ω ±5%, 1 W.

* FIGURE 18. Maximum clock fall time and threshold clock input test circuit for table XIV.

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NOTES:
1. Input pulse parameters for maximum clock fall time test:
   PW = 1.5 ±0.05 µs.
   t_{TLH} = 500 ±50 ns.
   t_{THL} = 1.0 ±.05 µs.
   V1 = 0 ±0.2 V.
   V2 = 3 ±0.1 V.
2. Input pulse parameters for threshold clock input test:
   PW = 200 ±10 ns.
   t_{TLH}, t_{THL} = 10 ±2 ns.
   V1 = 0.6 ±.05 V.
   V2 = 2.0 ±.05 V.
3. Before making measurements, the output levels shall be verified to be 0.3 V maximum for the low level and 2.4 V minimum for the high level.
4. All voltages are referenced to ground.

* FIGURE 19. Maximum clock fall time and threshold clock input test circuit waveforms for table XIV.

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TABLE XV. Output crosstalk (low-level and high-level) test.

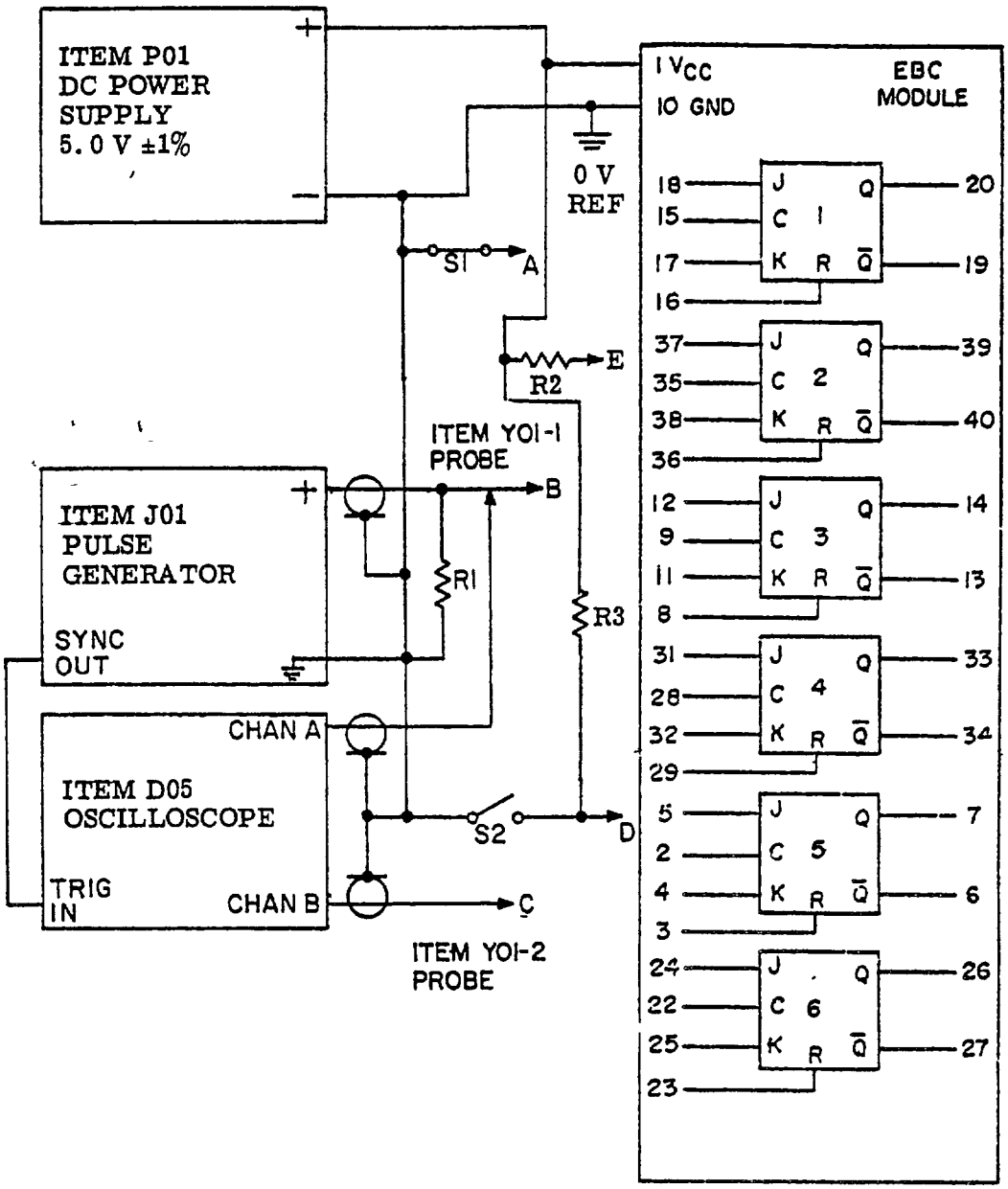
Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
W1	Q1 Output crosstalk, low level	15,16	2,9,22,28,35	20	36,8,29,3,23	Open	0	400	mV	20,21
W2	Q2	35,36	15,9,28,2,22	39	16,8,29,3,23					
W3	Q3	8,9	15,35,28,2,22	14	16,36,29,3,23					
W4	Q4	28,29	15,35,9,2,22	33	16,36,8,3,23					
W5	Q5	2,3	15,35,9,28,22	7	16,36,8,29,23					
W6	Q6	22,23	15,35,9,28,2	26	16,36,8,29,3					
W7	$\overline{Q}1$	15	35,9,28,2,22	19	17,38,11,23, 29,32,4,25, 36,3,8	16				
W8	$\overline{Q}2$	35	15,9,28,2,22	40	17,38,11,32, 4,25,16,8,29, 3,23	36				
W9	$\overline{Q}3$	9	15,35,28,2,22	13	17,38,11,32, 4,25,16,3,29, 23,36	8				
W10	$\overline{Q}4$	28	15,35,9,2,22	34	17,38,11,32, 4,25,16,3,23, 36,8	29				
W11	$\overline{Q}5$	2	15,35,9,28,22	6	17,38,11,32, 4,25,16,36, 29,8,23	3				
W12	$\overline{Q}6$	22	15,35,9,28,2	27	17,38,11,32, 4,25,16,36, 8,29,3	23				

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TABLE XV. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
X1	$\overline{Q1}$ Output crosstalk, high level	15,16	35,9,28,2,22	19	36,8,29,3,23	Open	0	400	mV	20,21
X2	$\overline{Q2}$	35,36	15,9,28,2,22	40	16,8,29,3,23					
X3	$\overline{Q3}$	8,9	15,35,28,2,22	13	16,36,29,3,23					
X4	$\overline{Q4}$	28,29	15,35,9,2,22	34	16,36,8,3,23					
X5	$\overline{Q5}$	2,3	15,35,9,28,22	6	16,36,8,29,23					
X6	$\overline{Q6}$	22,23	15,35,9,28,2	27	16,36,8,29,23					
X7	Q1	15	35,9,28,2,22	20	17,38,11,36,8,29,32,4,25,3,23	16				
X8	Q2	35	15,9,28,2,22	39	17,38,11,16,8,29,32,4,25,3,23	36				
X9	Q3	9	15,35,28,2,22	14	17,38,11,16,36,32,4,25,29,3,23	8				
X10	Q4	28	15,35,9,2,22	33	17,38,11,16,36,32,4,25,8,3,23	29				
X11	Q5	2	15,35,9,28,22	7	17,38,11,16,36,32,4,25,8,29,23	3				
X12	Q6	22	15,35,9,28,2	26	17,38,11,16,36,32,4,25,8,29,3	23				

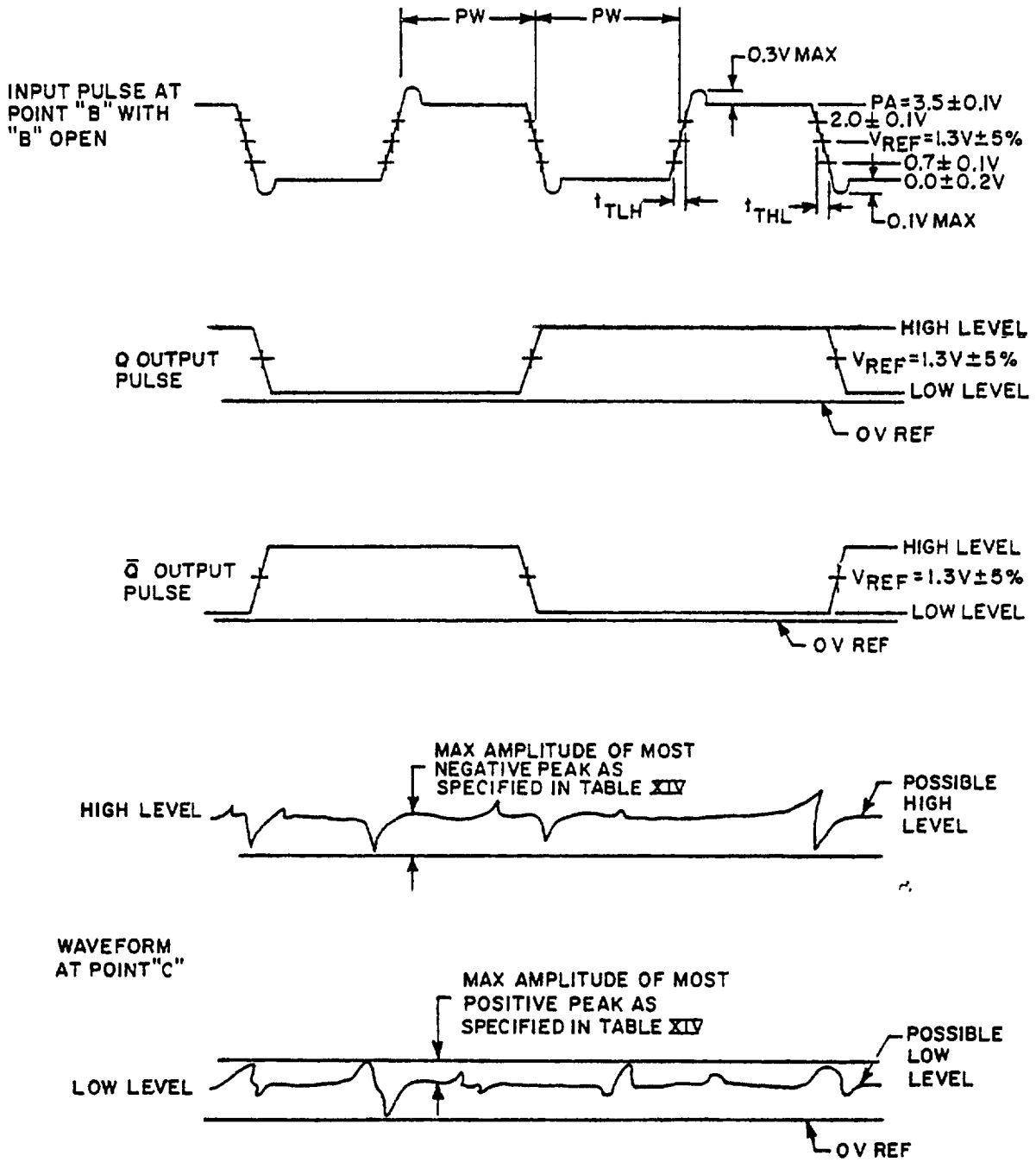
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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. R2 = R3 = 2.75 kΩ ±1%, 1/8 W.
3. S1, S2 = SPST switch.
4. Switching sequence for conditions W1-W6, and X1-X6:
   a. Switch S1 closed.
   b. Close switch S2 for a minimum Of 500 ns, then open and make measurement.
5. Switching sequence for test condition W7-W12 and X7-X12:
   a. S1 and S2 closed.
   b. Open switch S1 for a minimum of 500 ns, then close it.
   c. Open switch S2 and make measurement.
6. Load all outputs with the load as specified on figure 10.

* FIGURE 20. Output crosstalk (low-level and high-level) test circuit for table XV.

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NOTE: Input pulse parameters:
PW = 500 ±50 ns.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at all inputs).

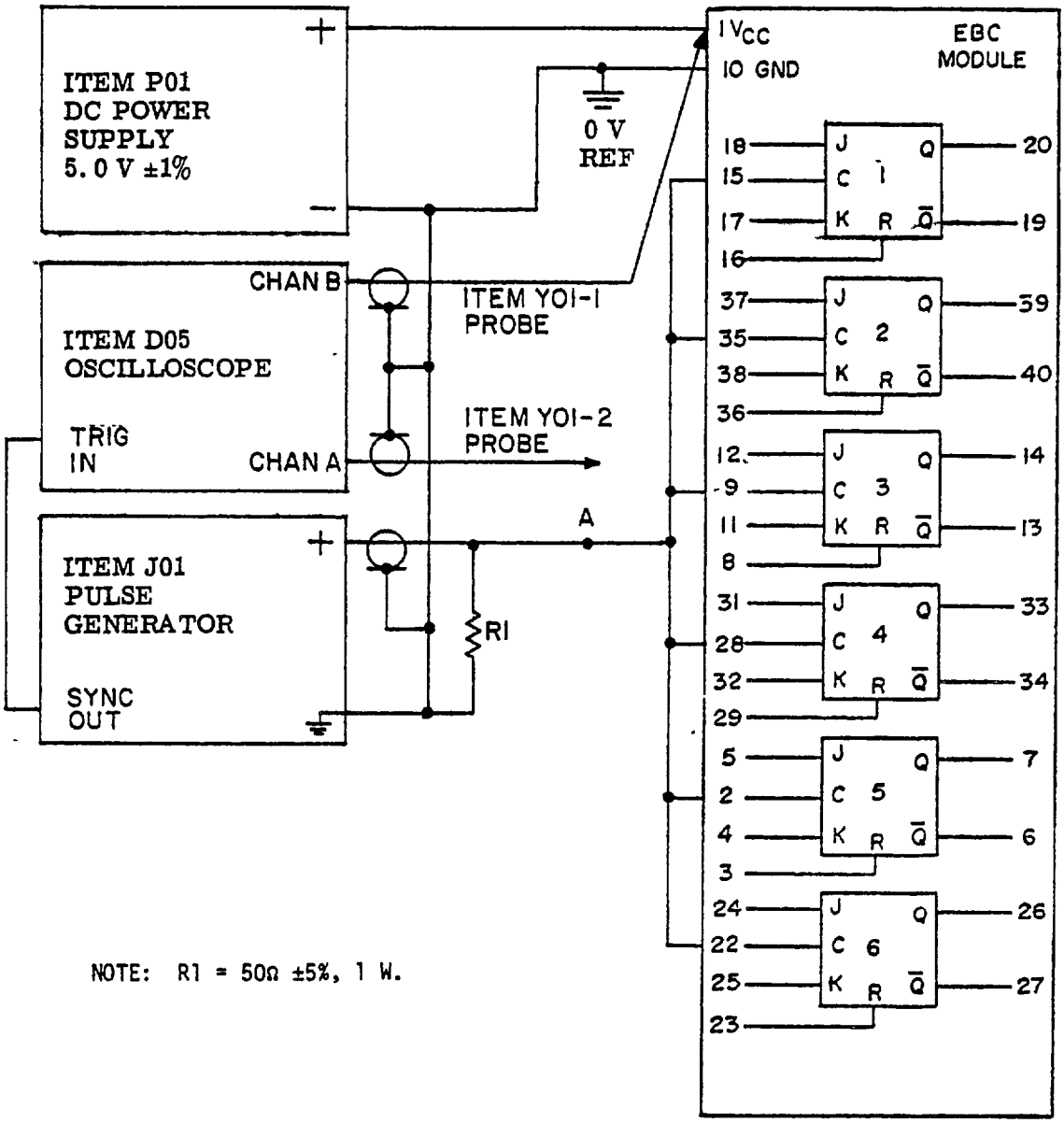
* FIGURE 21. Output crosstalk (low-level and high-level) test circuit waveforms for table XV.

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TABLE XVI. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
Y	Power supply transient amplitude	0	400	mV	22,23

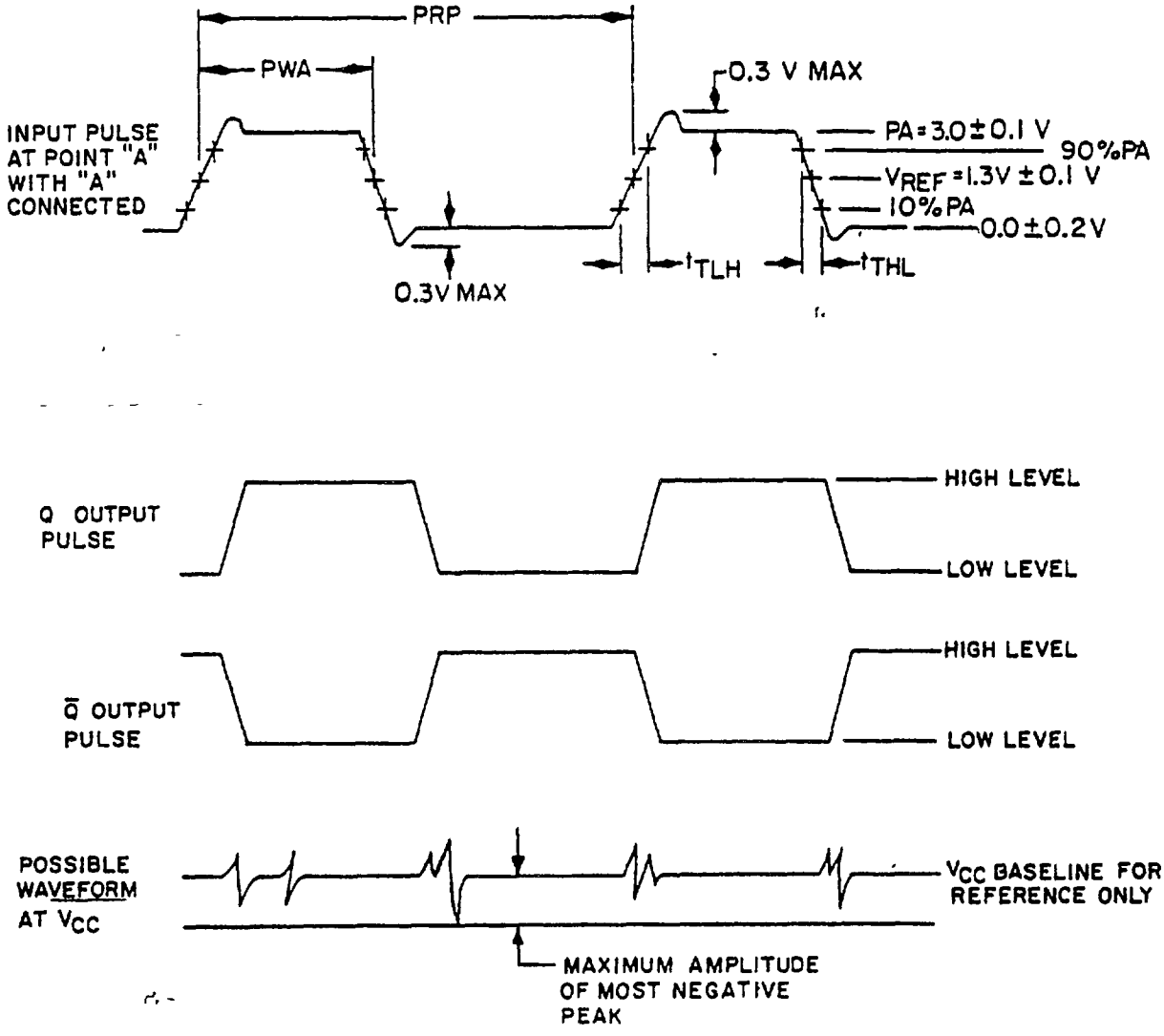
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NOTE: R1 = 50Ω ±5%, 1 W.

* FIGURE 22. Power supply transient amplitude test circuit for table XVI.

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NOTE: Input pulse parameters:
$t_{TLH} = t_{THL} = 20 \pm 2$ ns.
PWA = 500 ± 50 ns.
PRP = 1,000 ±100 ns.

* FIGURE 23. Power supply transient amplitude test circuit waveforms for table XVI.

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* TABLE XVII. Power supply filter capacitance test.

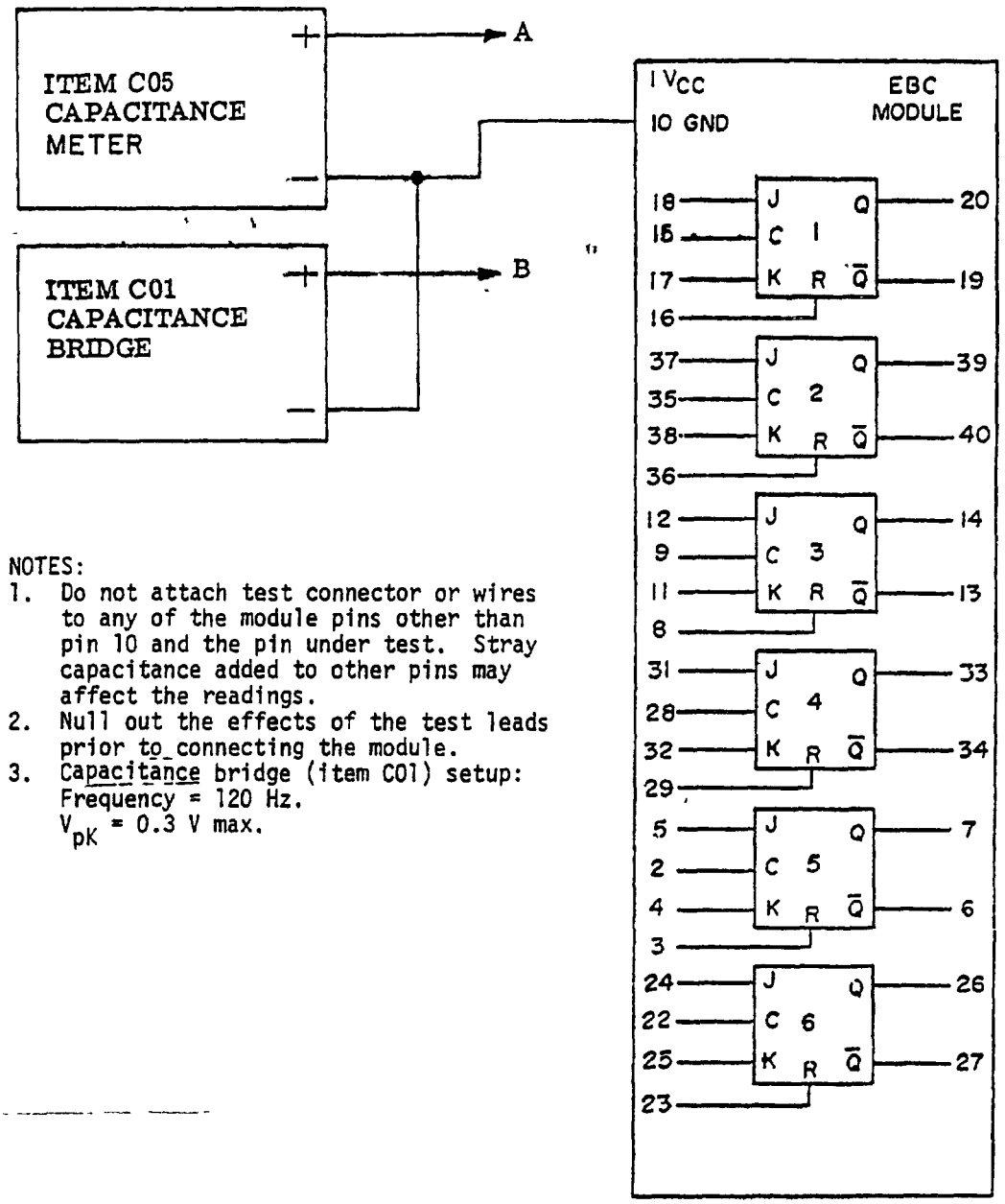
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Z	Power supply filter capacitance	Open	1	1.65	2.75	µF	24

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TABLE XVIII. Input capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
AA1	C1 Input capacitance	15	Open	2	15	pF	24
AA2	C2	35					
AA3	C3	9					
AA4	C4	28					
AA5	C5	2					
AA6	C6	22					
AA7	J1	18					
AA8	J2	37					
AA9	J3	12					
AA10	J4	31					
AA11	J5	5					
AA12	J6	24					
AA13	K1	17					
AA14	K2	38					
AA15	K3	11					
AA16	K4	32					
AA17	K5	4					
AA18	K6	25					
AA19	R1	16					
AA20	R2	36					
AA21	R3	8					
AA22	R4	29					
AA23	R5	3					
AA24	R6	23					

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NOTES:
1. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
2. Null out the effects of the test leads prior to connecting the module.
3. Capacitance bridge (item C01) setup: Frequency = 120 Hz.
   V_{pK} = 0.3 V max.

* FIGURE 24. Power supply filter capacitance and input capacitance test circuit for tables XVI and XVII.

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